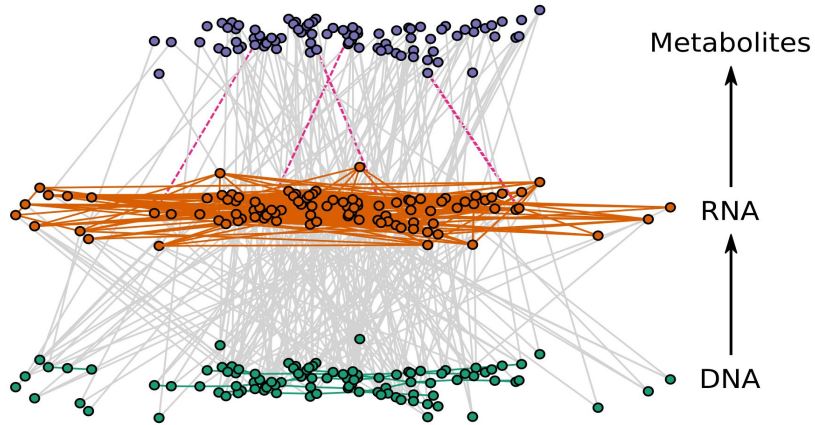


Lecture 0: Introduction to Applied Research in Health Data Science

CSCI6410/4148 & EPAH6410

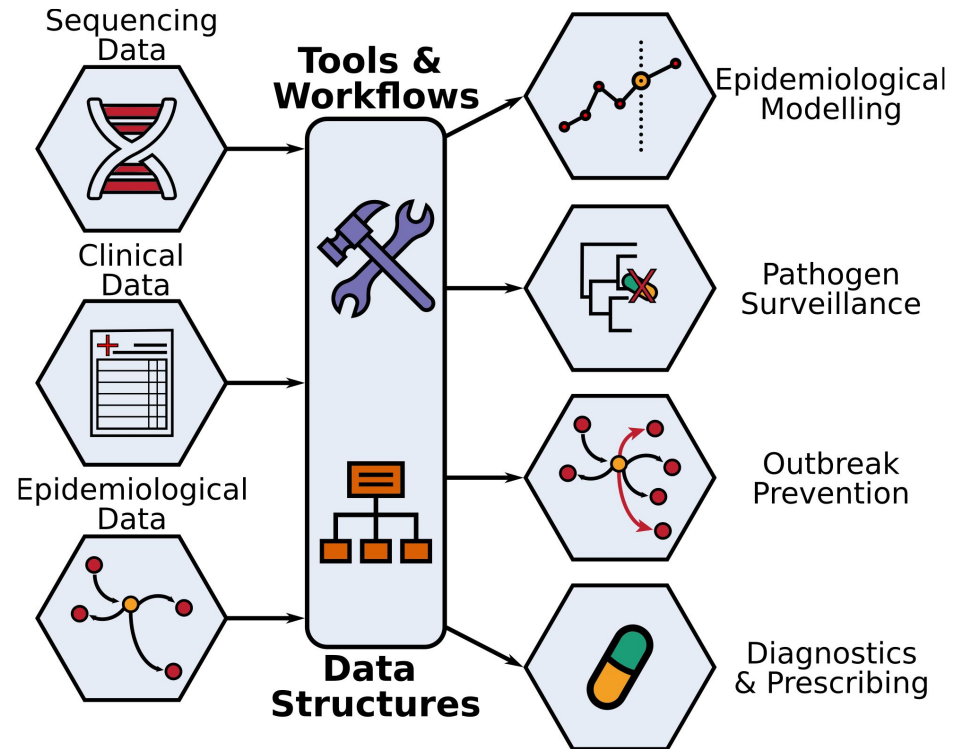
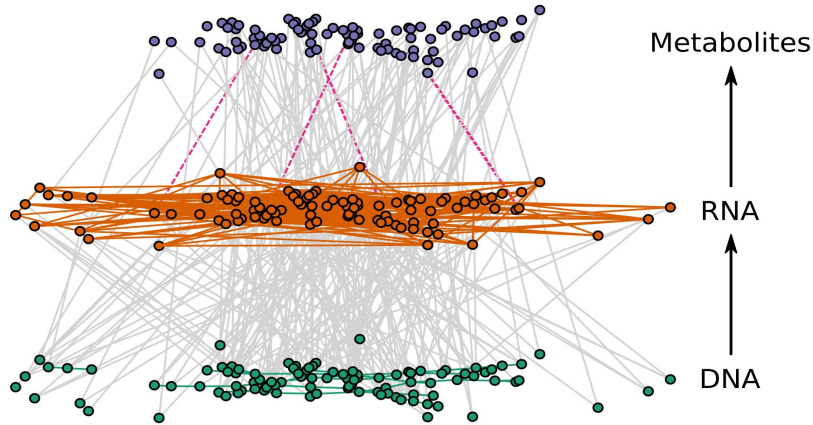
Finlay Maguire (finlay.maguire@dal.ca)
TA: Mehrana Calagari (m.calagari@dal.ca)

Why am I teaching this course?



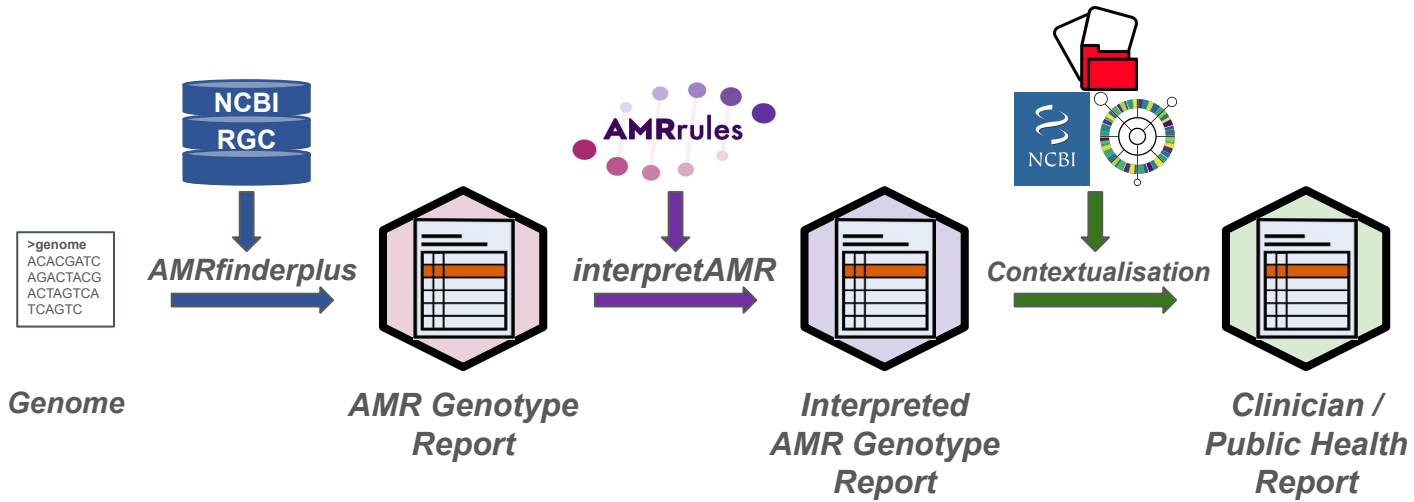
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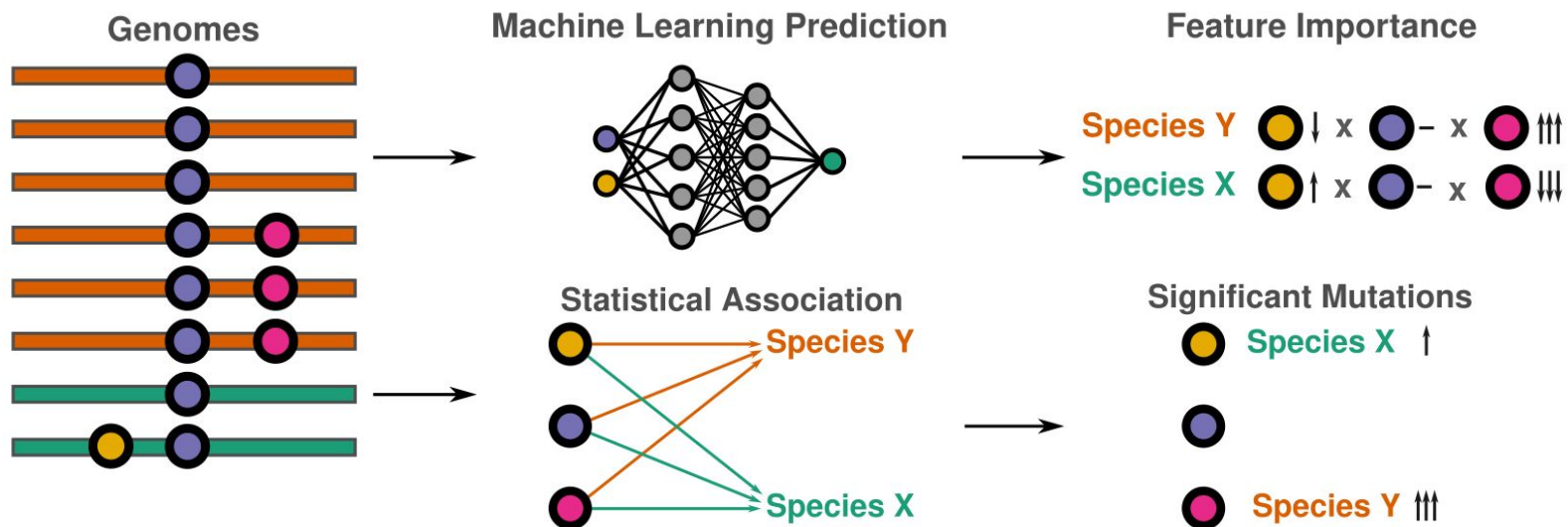
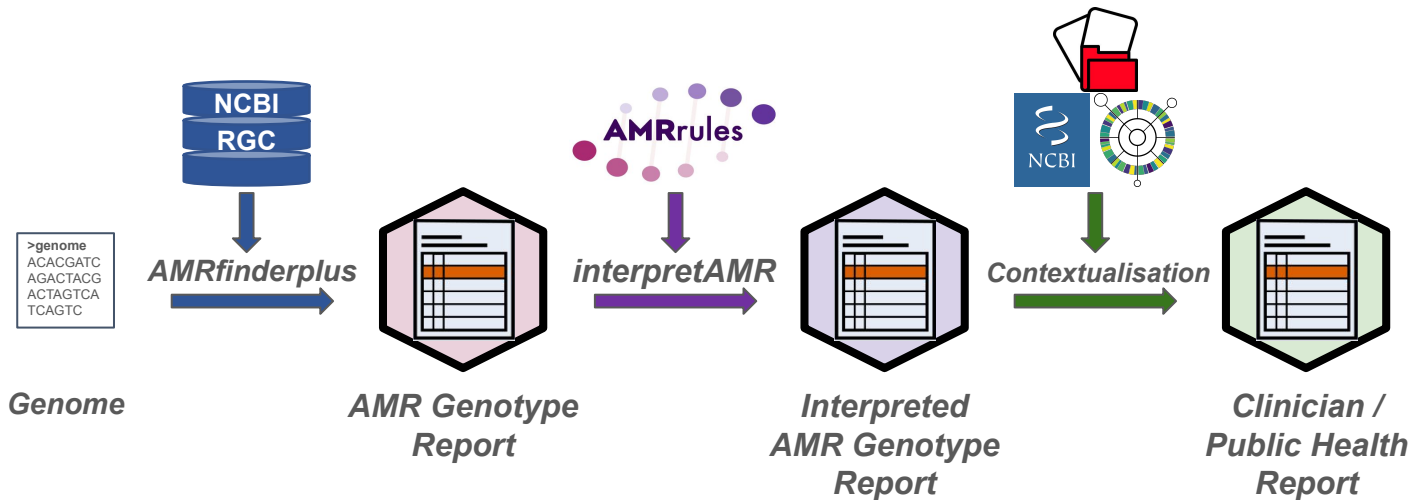


- **PhD (Bioinformatics):** using large noisy datasets to understand how microbial systems and mechanisms evolve.
- **Research Group (Genomic Epidemiology):** using large noisy datasets to better diagnose, track and predict infectious diseases.

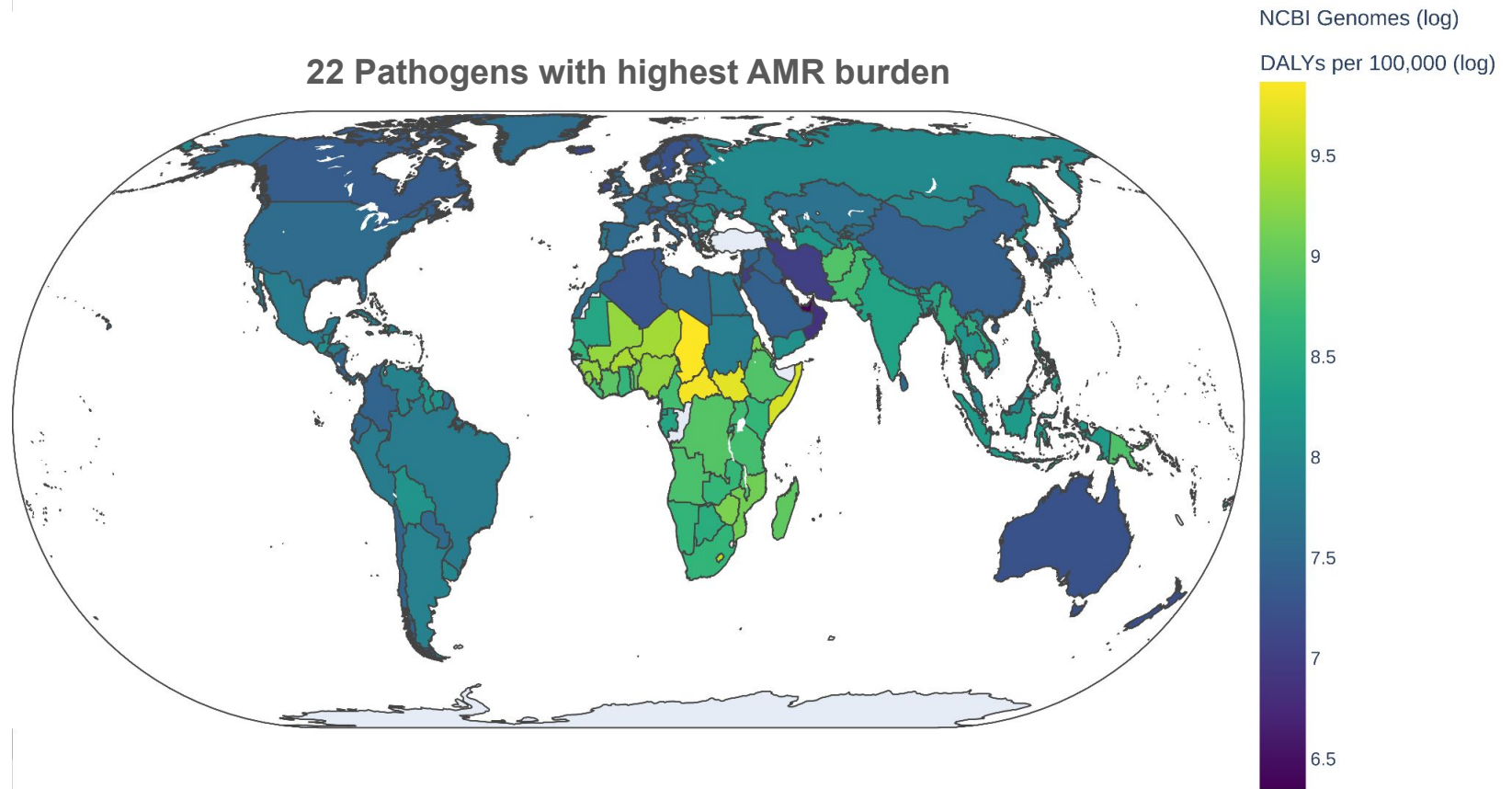
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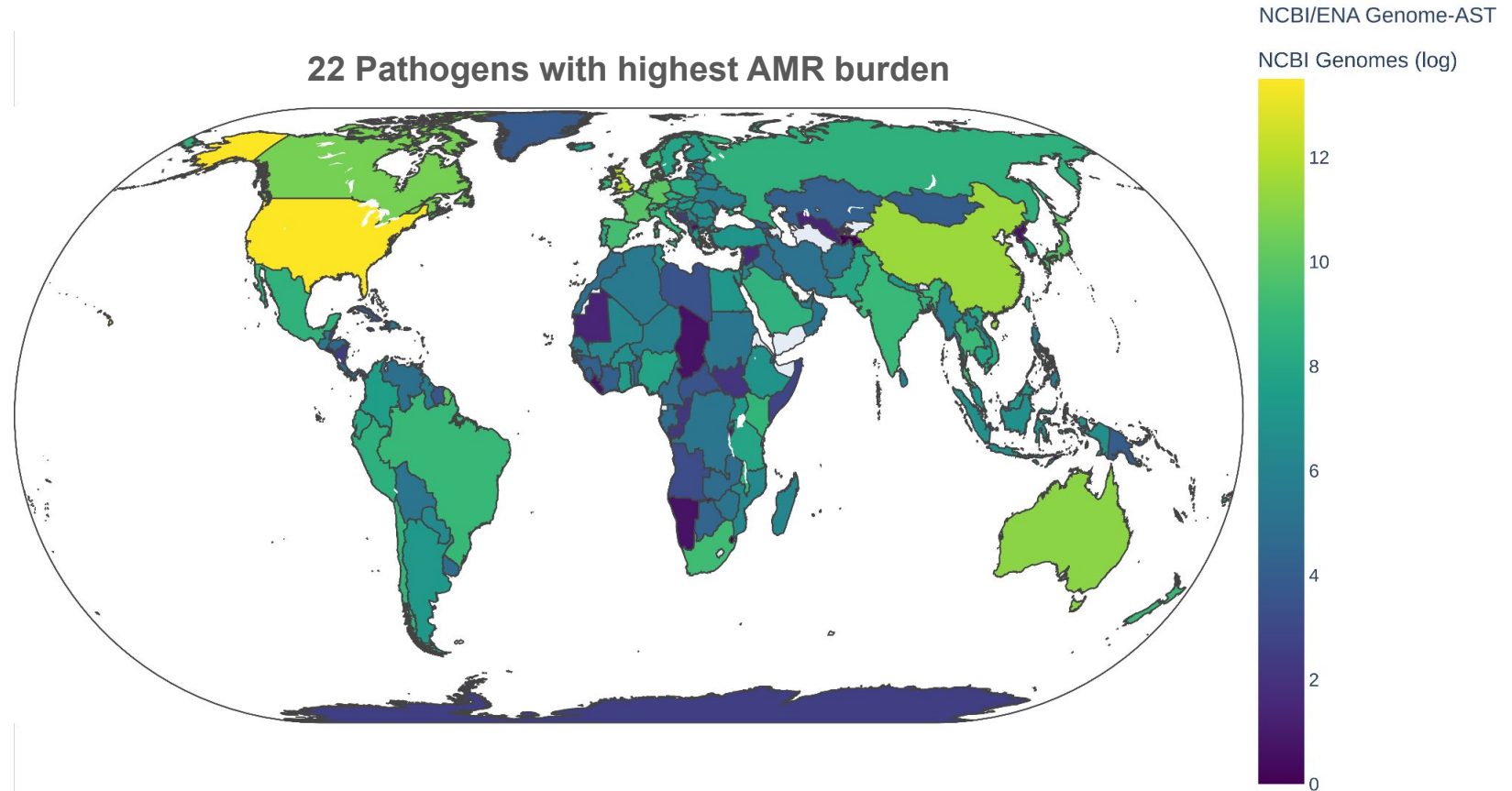


Why am I teaching this course?



K. pneumoniae: 23,379,605 DALYs across 100 highest burden countries

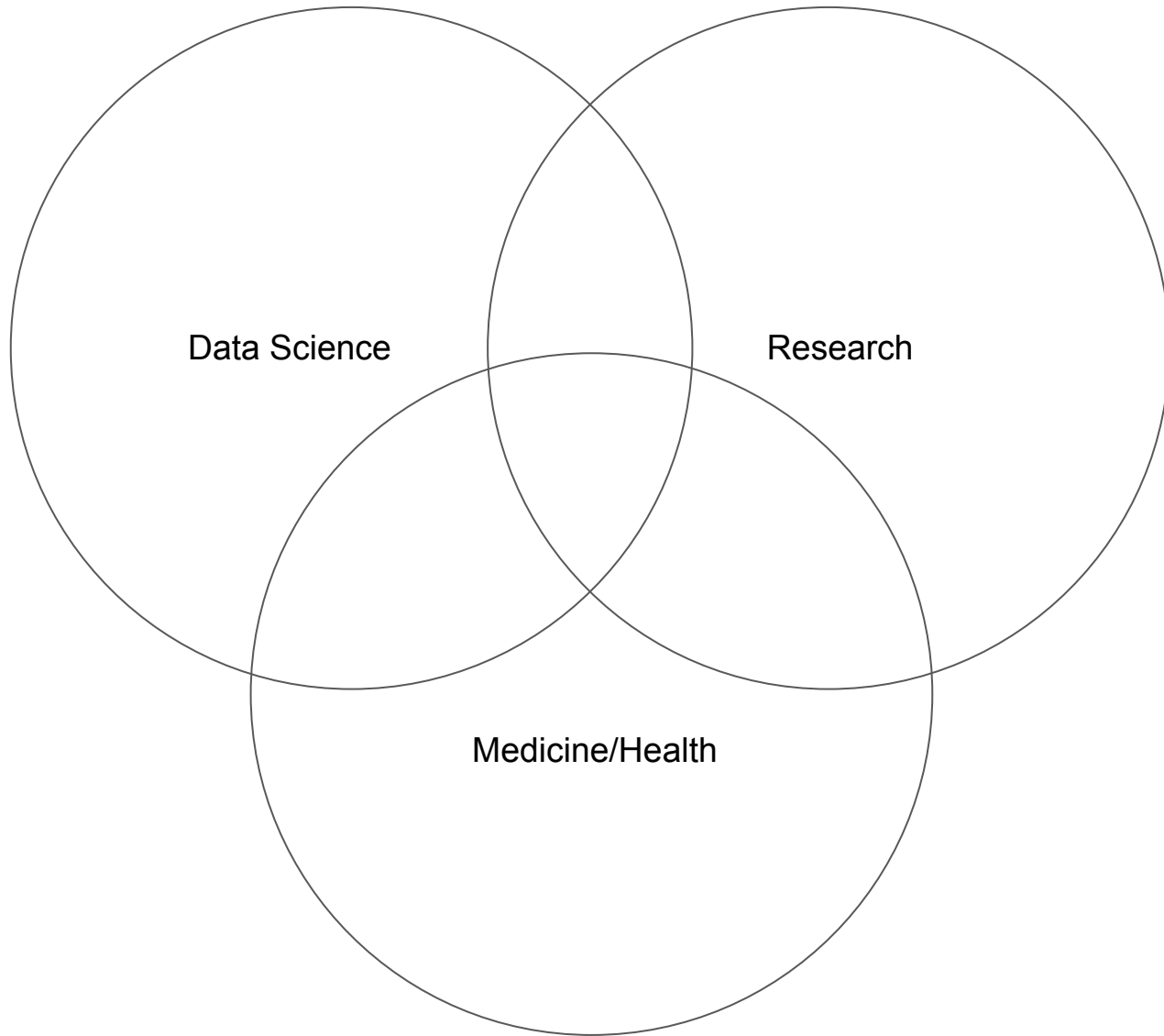
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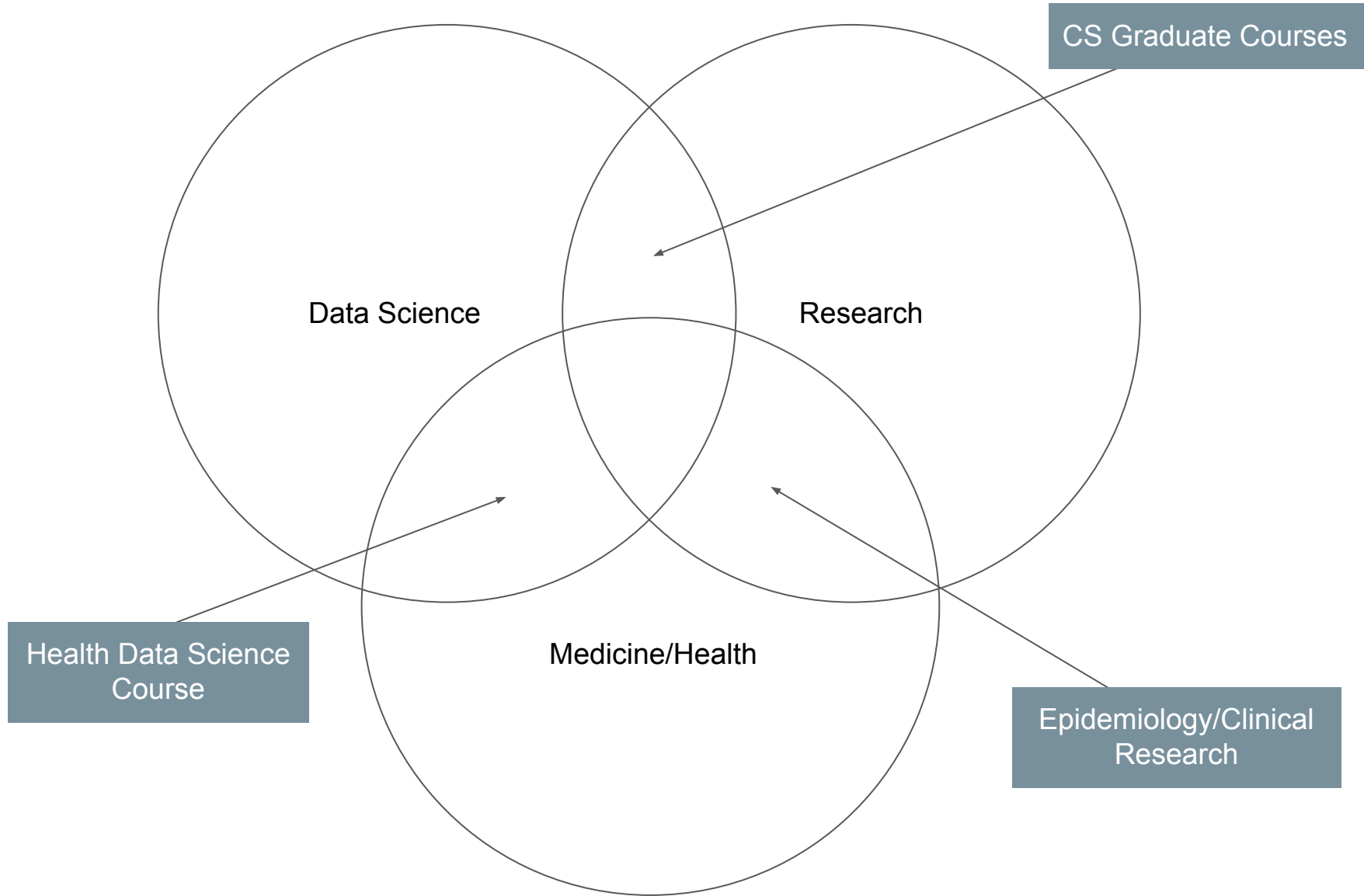
K. pneumoniae: 8,689/89,056 genomes from 100 highest burden countries

Overview of course

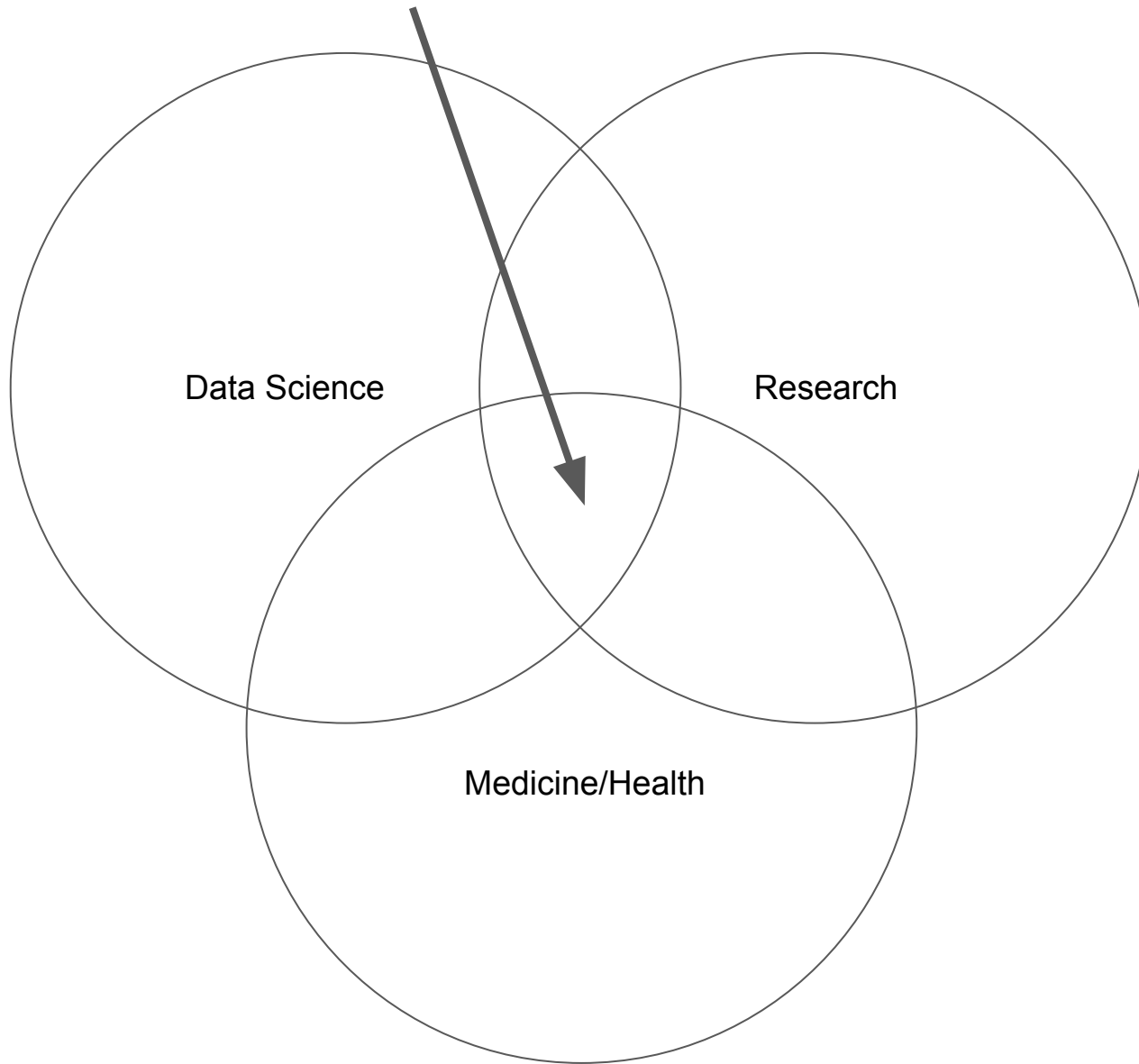
Applied Research in Health Data Science



Applied Research in Health Data Science



Applied Research in Health Data Science



Learning Outcomes

1. Understand the **4 principal sources and data types** of medical data:
 - a. longitudinal databases (tabular)
 - b. electronic medical records (structured, semi-structured, and unstructured text)
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 - d. physiological (signal and time-series).

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6. Combine these skills to develop high-quality collaborative health data science **research proposals**

What is not covered in this course

- **Breadth/depth** of each data science method: *each could be multiple graduate CS courses*

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- True **messiness** of real data: *provide tools but experience is invaluable*
- Some important forms of medical data (e.g., genomics): *see CSCI4181/6810, EPAH6052, come speak to me if interested in this specifically.*

Course Structure

Overview of data types & analysis methods:

- **Lectures** (Monday/Wednesday)

Course Structure

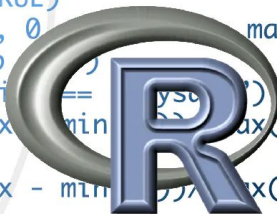
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(CSCI4148: drop lowest scoring assignment)

```
dens <- density(data, n = npts)
dx <- dens$x
dy <- dens$y
if(add == TRUE)
  plot(0., 0., main = "Density Plot",
       ylab = "Density",
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         dx2 <- (dx - min(dx)) / max(dx)
         x[1.]
         dy2 <- (dy - min(dy)) / max(dy)
         y[1.]
         seqbelow <- rep(y[1.], length(dx))
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<https://www.coursera.org/learn/r-programming>

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2 papers per week, randomly assigned rota for leading discussion of paper with rest of class.

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Paper presentation (20%)

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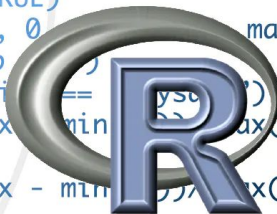
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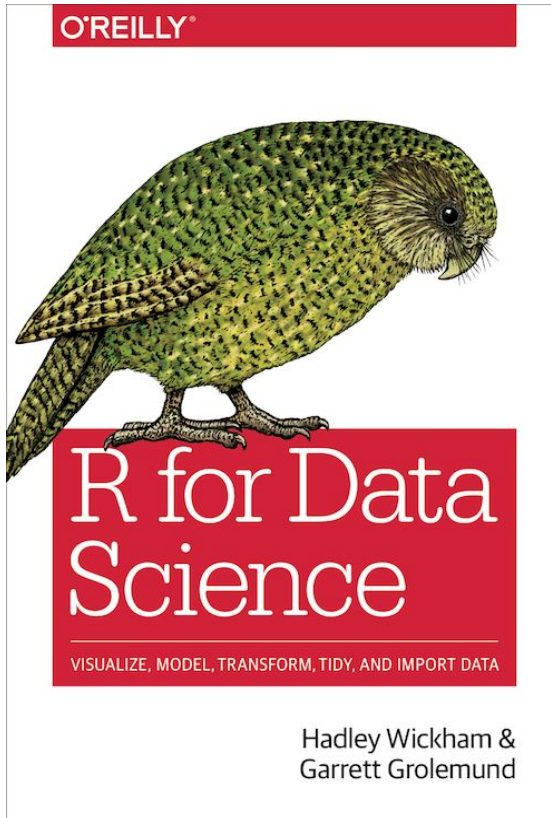
- **Class** (Wednesday/Friday)

Assessment:

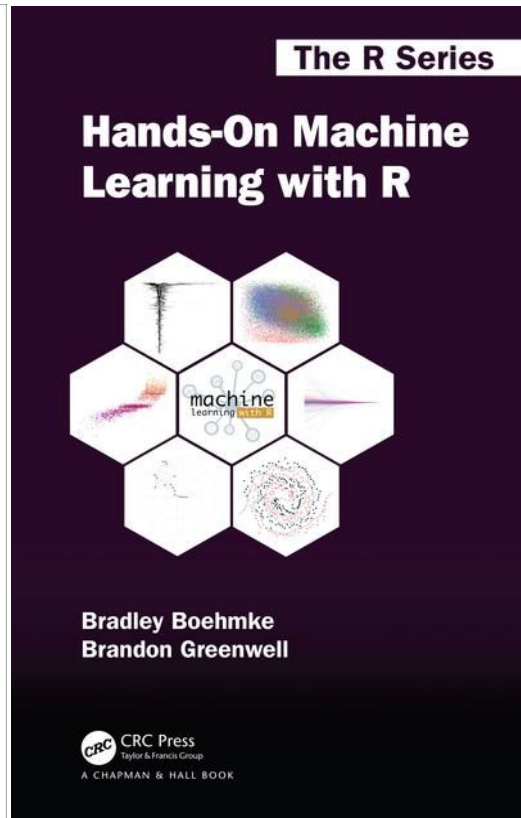
Presentation **last full week of class** (20%)

Submitted **final day of class** (20%)

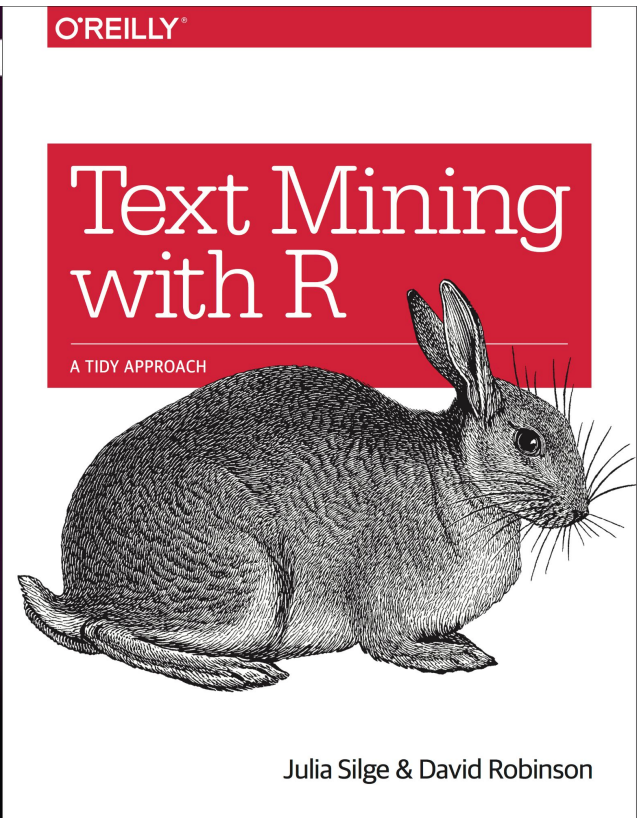
Course Materials



<https://r4ds.had.co.nz/>



<https://bradleyboehmke.github.io/HOML/>



<https://www.tidytextmining.com/>

Course Website



CSCI6410/CSCI4148/EPAH6410: Applied Research in Health Data Science / Summer
2025-2026

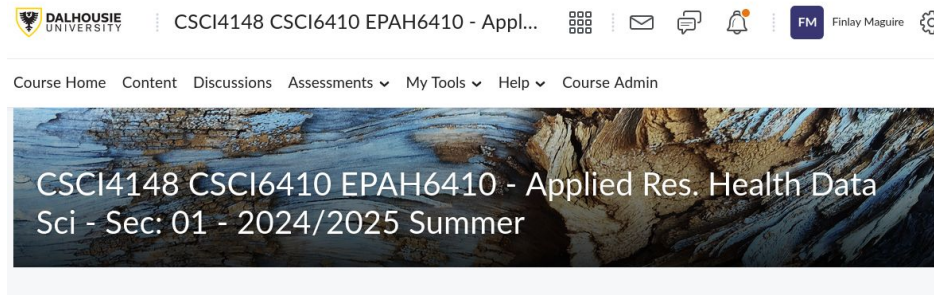
https://maguire-lab.github.io/health_data_science_research_2026/

Course Website



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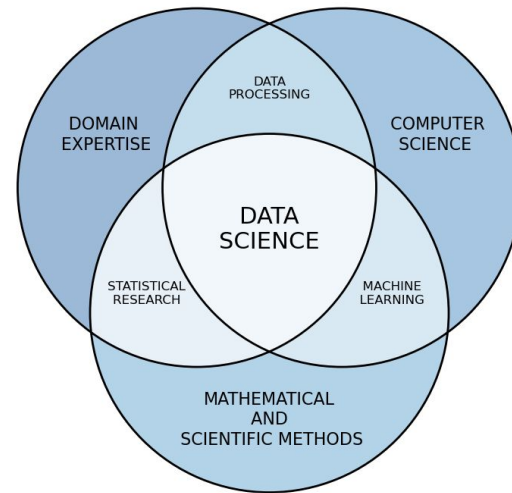


Grades/Submissions:

<https://dal.brightspace.com/d2l/home/385844>

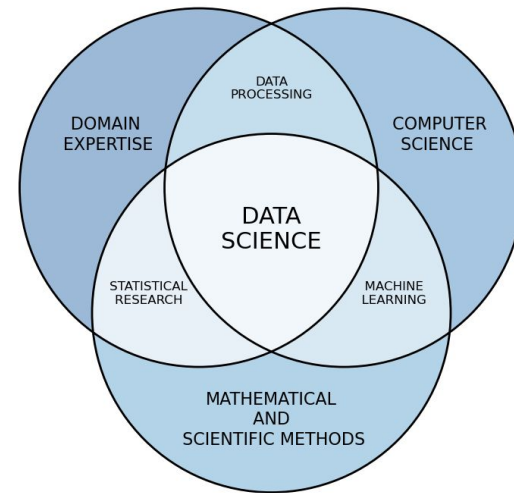
What is ~~health~~ data science?

Data Science: *Data-intensive interdisciplinary approaches to understand and predict with secondary/live data*



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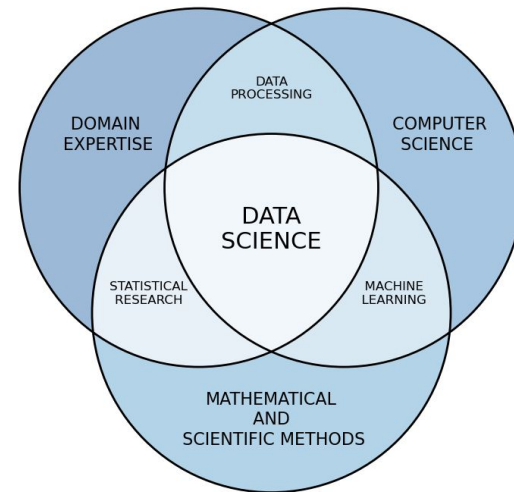
A range of partial and totally overlapping terms:



Data Science: *Data-intensive interdisciplinary approaches to understand and predict with secondary/live data*

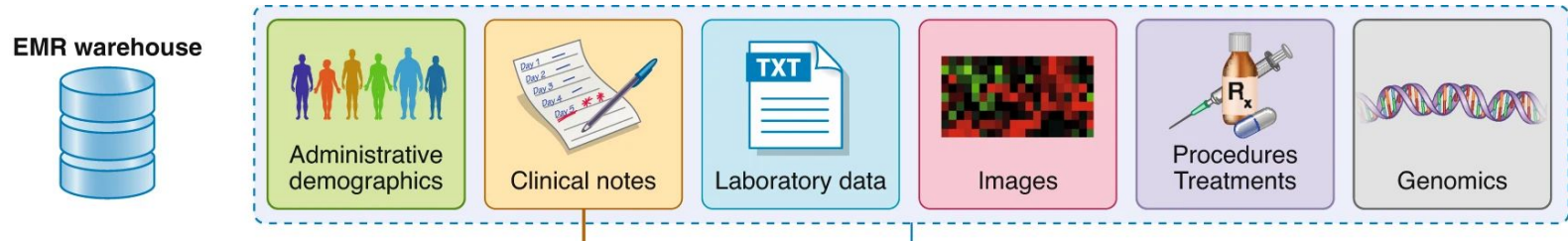
A range of partial and totally overlapping terms:

- Data Analytics
- Data Engineering
- Data Mining
- {Health,Bio,Medical}Informatics
- Database Analysis
- Business Intelligence
- Epidemiology
- Statistics
- **Machine Learning**
- Pattern Recognition
- Predictive Analytics
- Quantitative Researcher
- Scientist
- Analyst
- Algorithmic Modeling



OK, what is **Health** Data Science?

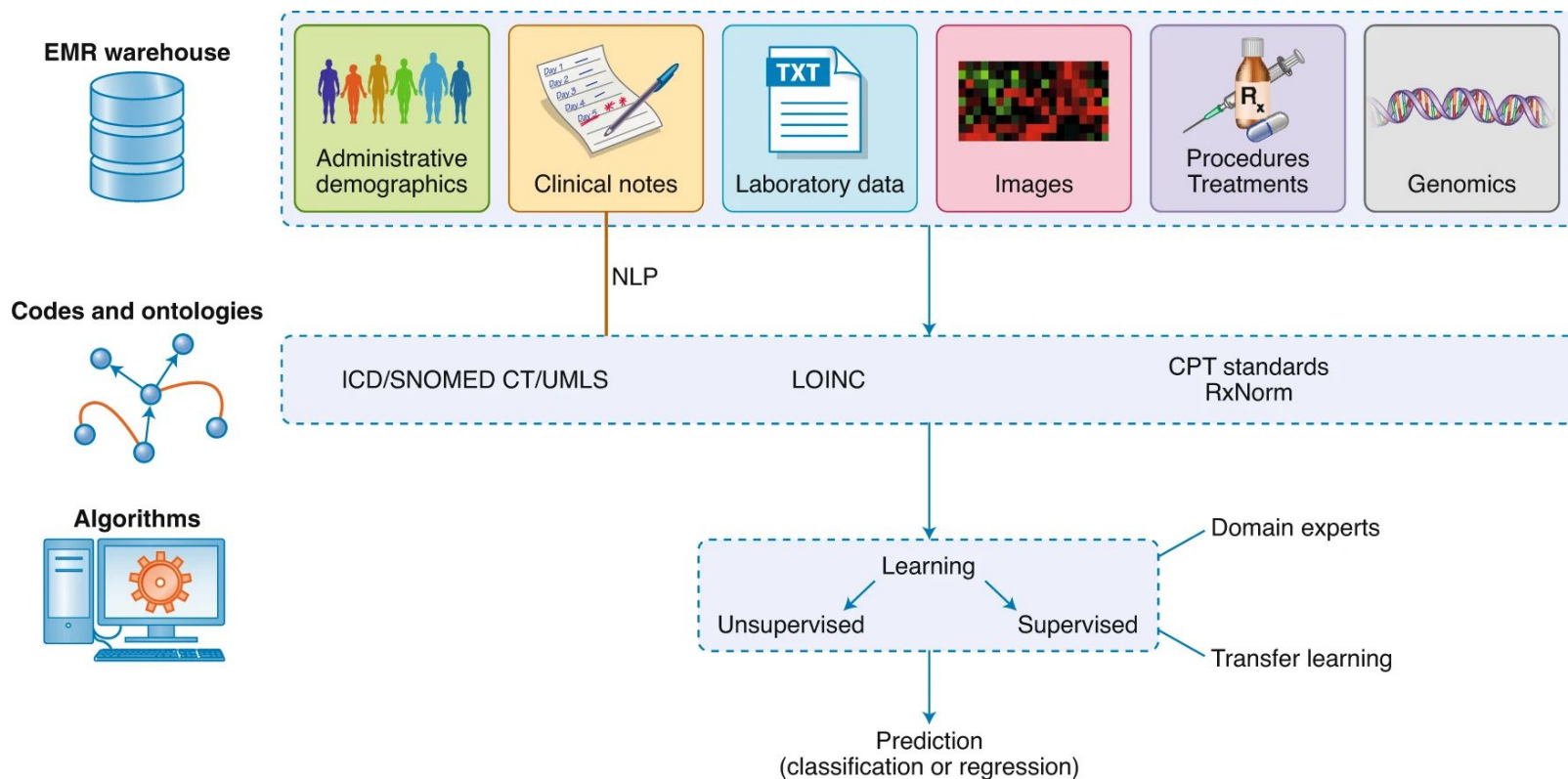
Data Science applied to Health Data



Why “health data” instead of “medical data”:

medical (being defined as MD-related”) $\subset$ health - **contentious**

Data Science applied to Health Data



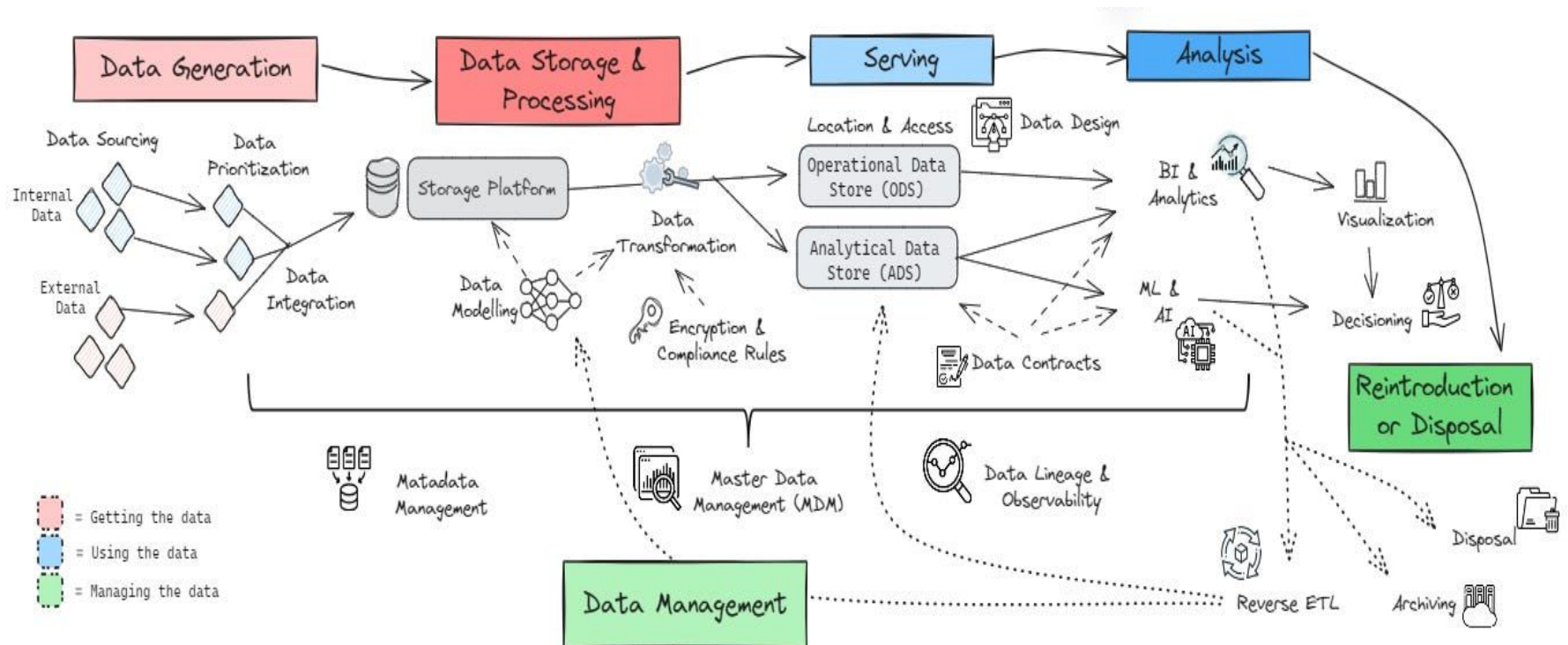
<https://www.nature.com/articles/s41588-020-0698-y/figures/2>

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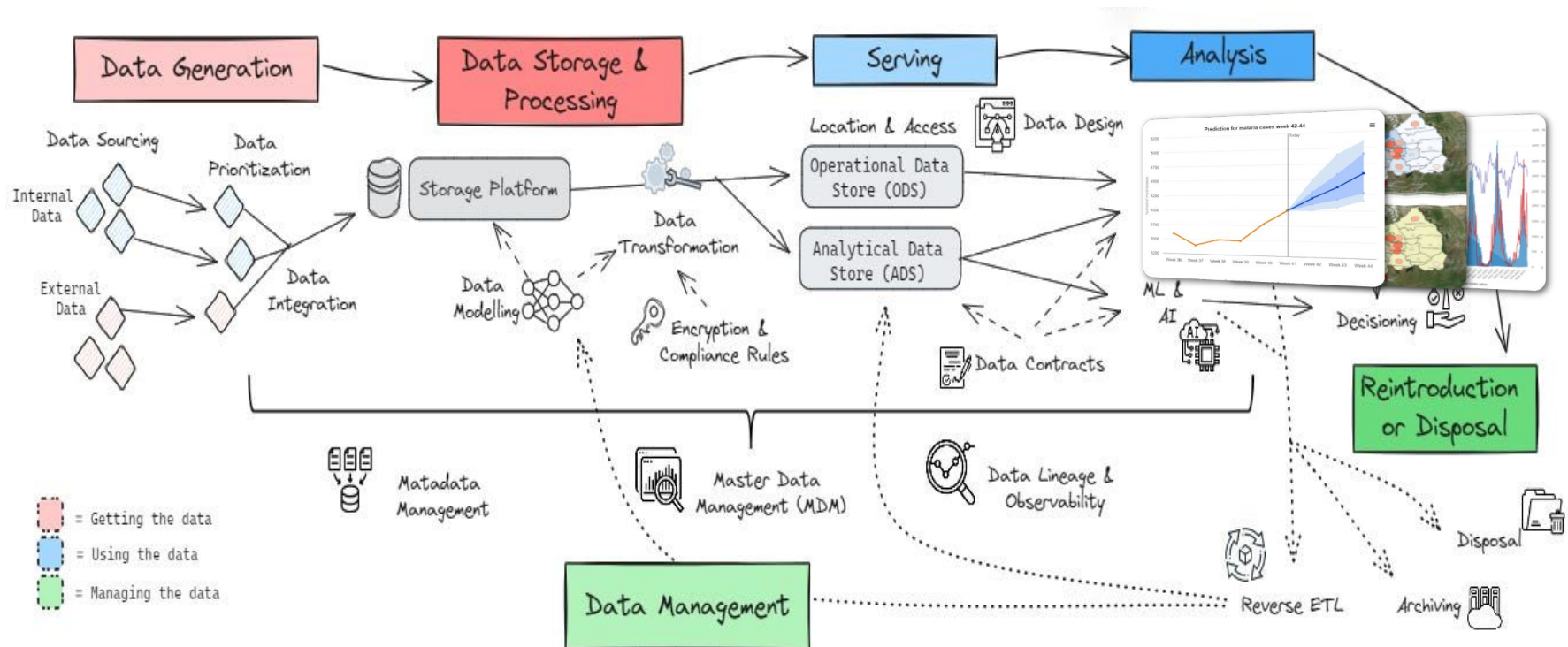
medical (being defined as MD-related”) $\subset$ health - **contentious**

How does (health) data science (generally)
differ from traditional epidemiology using
secondary data?

Data science integrates within the wider data ecosystem



Data science integrates within the wider data ecosystem



Important as datafication pervades medicine (and more)

Medical Notes:

- 1-50MB

Laboratory Values:

- 10-2000MB

Physiological Sensors:

- 0.1-200GB

Genomics:

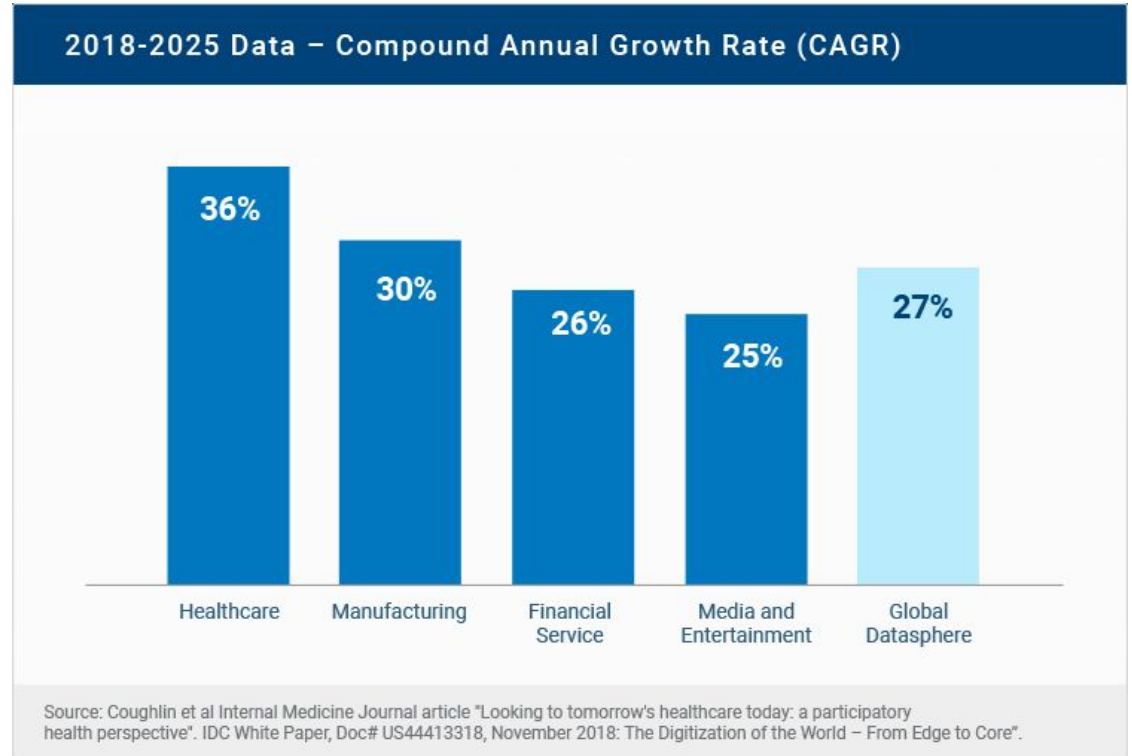
- 25-250GB

Medical Imaging:

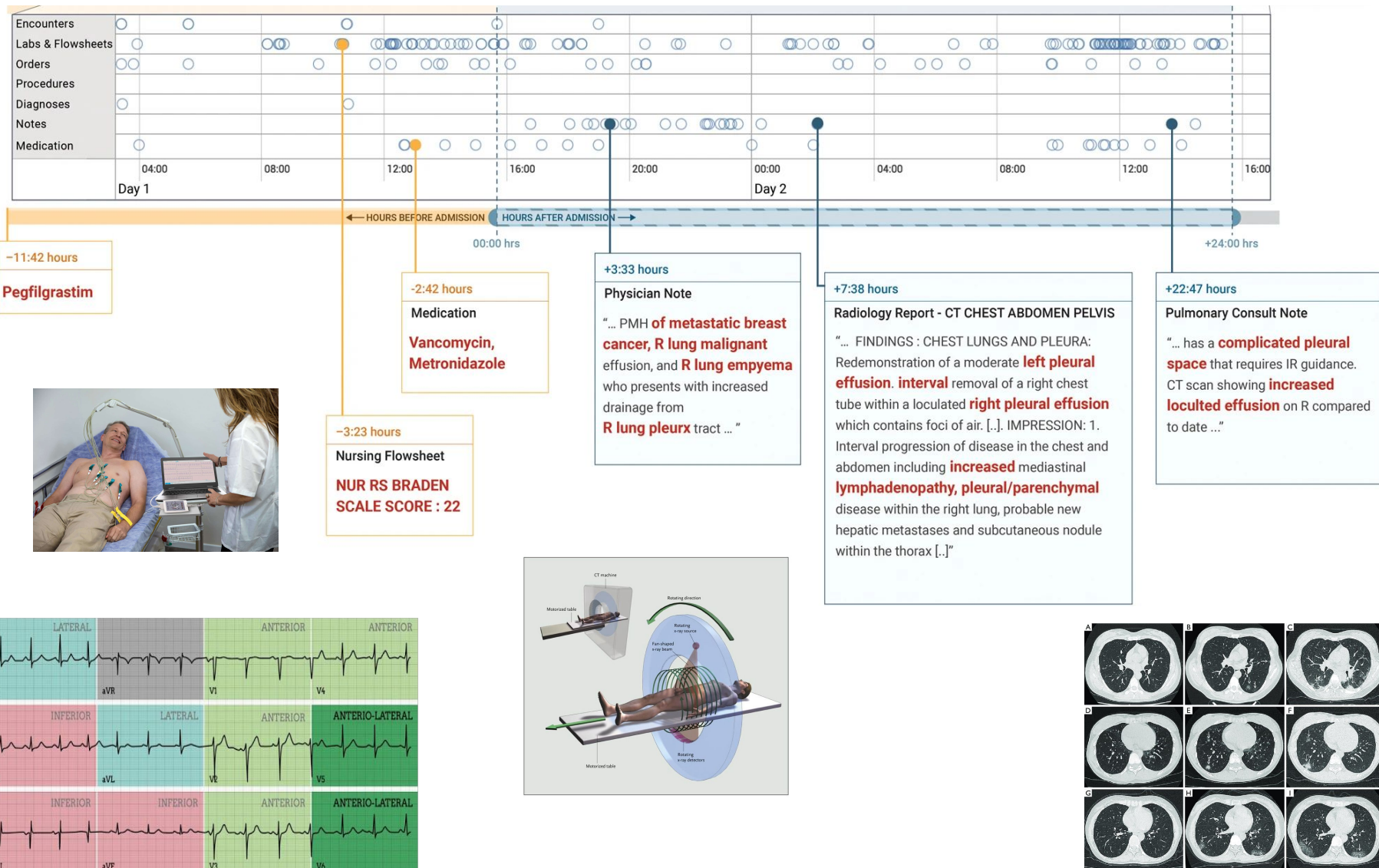
- 0.1-10TB

Hospital:

- 0.5-50PB per day



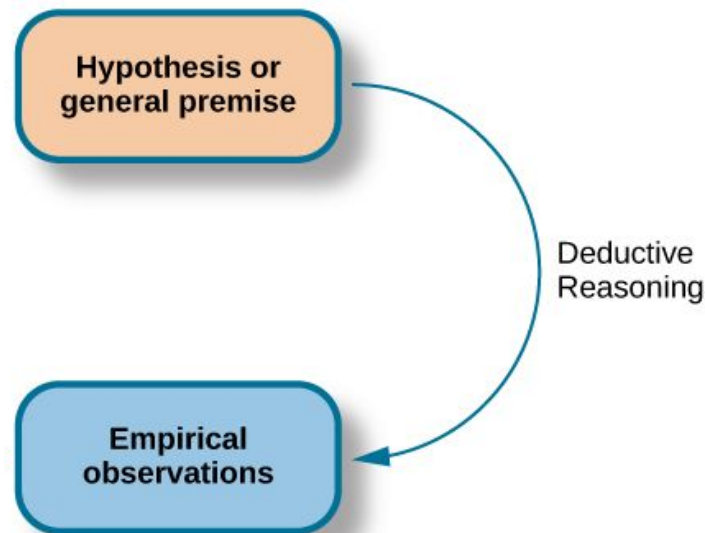
Data Science handles unstructured and multi-modal data



Data science supports inductive approaches

Deductive:

- “Condition X, causes Y”
- Collect data
- Perform (typically) frequentist statistical tests
- Reject or confirm null hypothesis



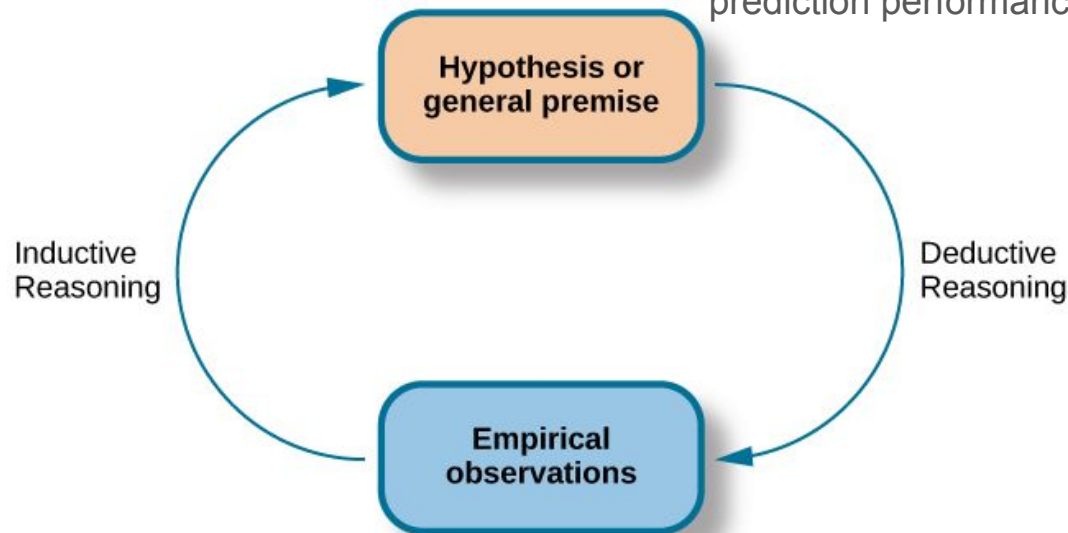
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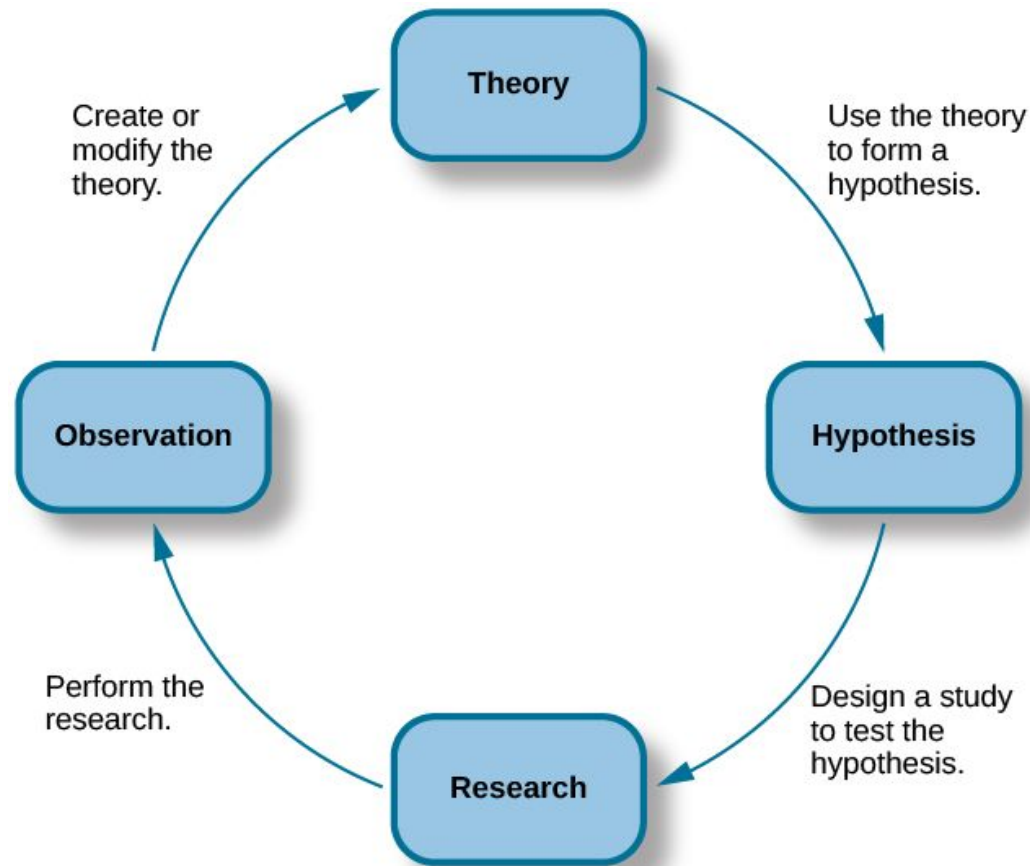
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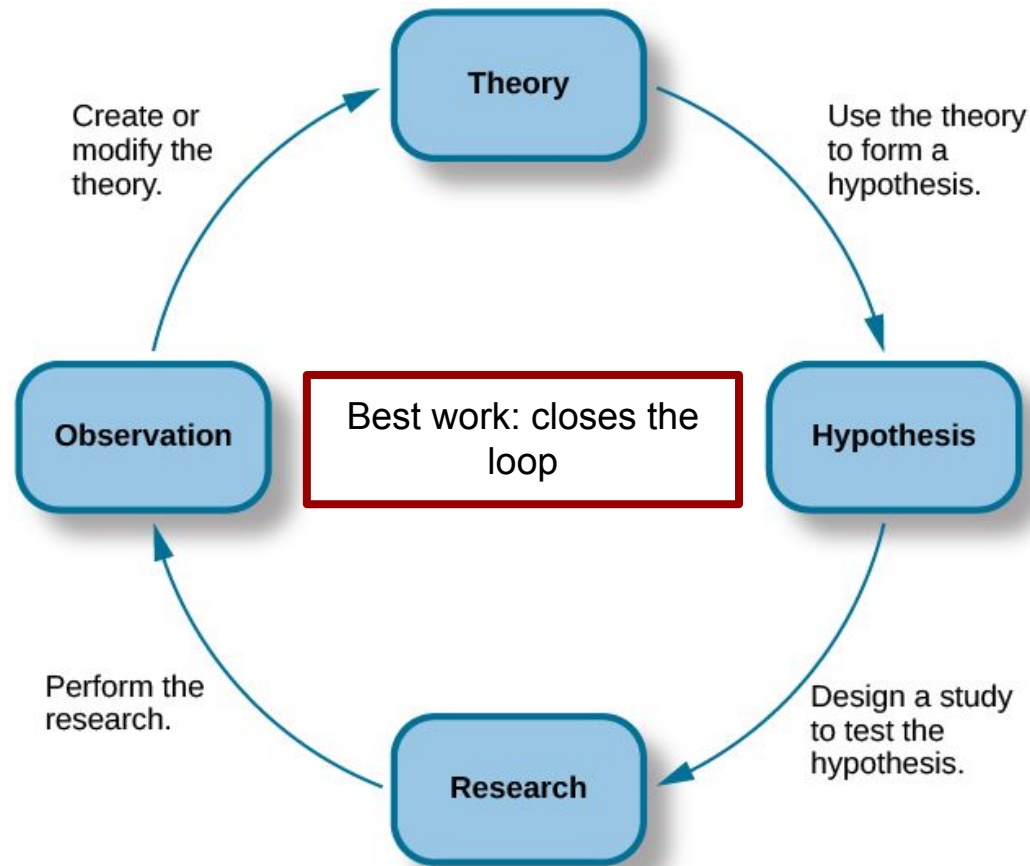
- Collect data
- Identify patterns in the data
- Observe X and Y seem connected somehow
- Quantify strength of association e.g., prediction performance



Data science aligns with knowledge cycle

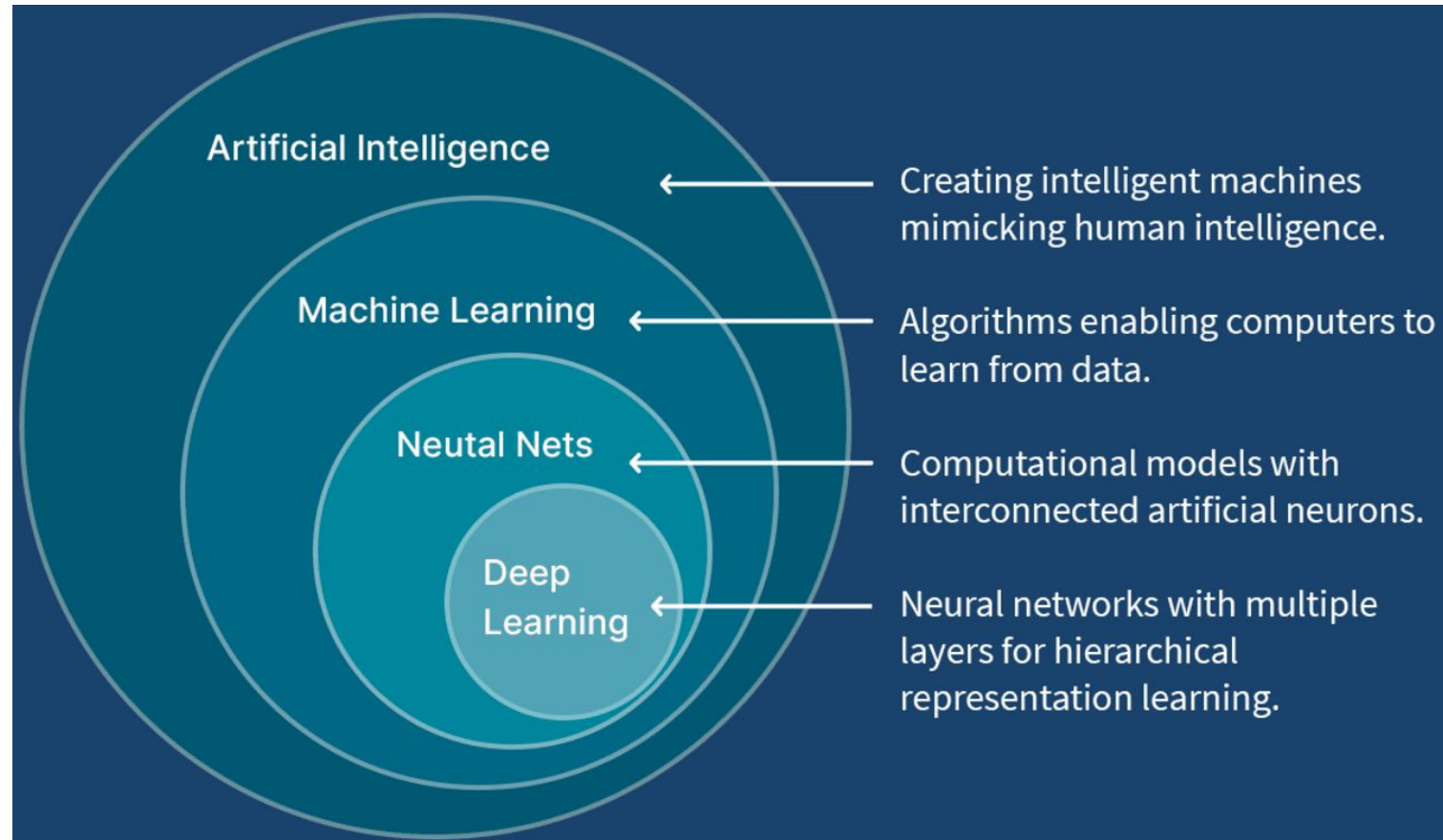


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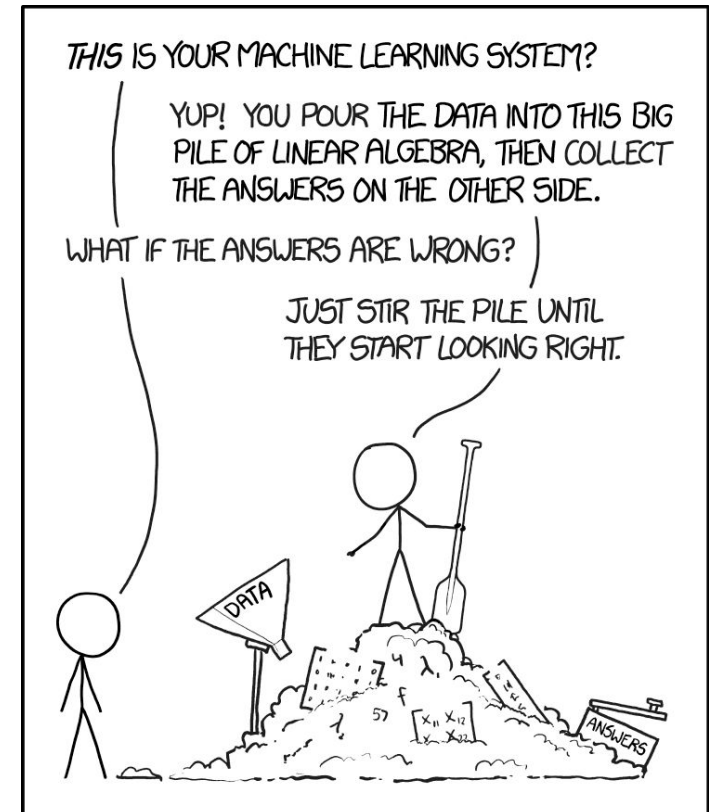
Most obvious difference: Data Science often
is based in Machine Learning

Machine Learning is a subset of Artificial Intelligence



ML involves finding and using patterns in data

- “Machine Learning is the field of study that gives the computer the ability to learn without being explicitly programmed”
- “A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P , if its performance at tasks in T , as measured by P , improves with experience E .”
- **Task** (*play checkers*)
- **Experience** (*data*):
 - games played by the program (with itself)
- **Performance** (*metric*):
 - How often does it win
- Training models which identify patterns in data



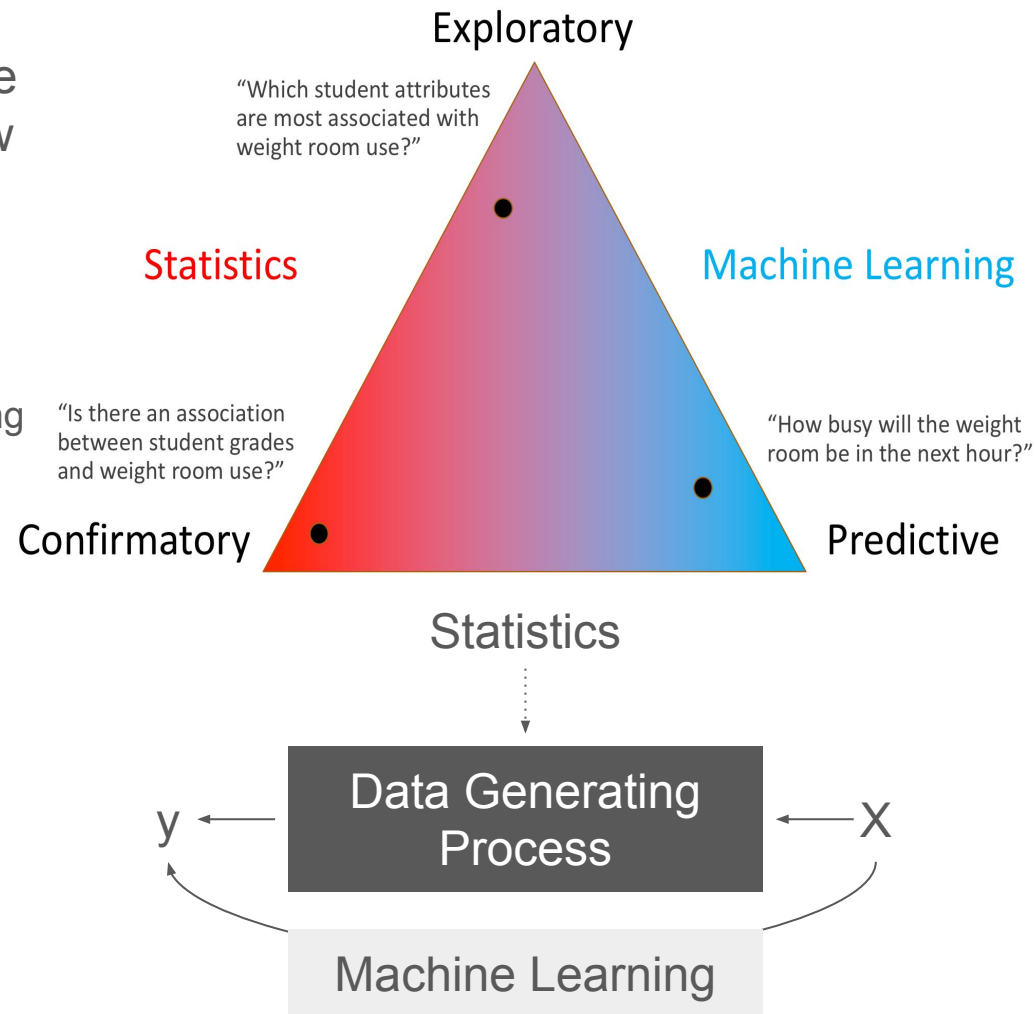
Is Machine Learning just a CS-flavoured
rebrand of statistics?

Large overlap but difference in people and priorities

- Many shared methods
- Difference in focus/priorities/culture

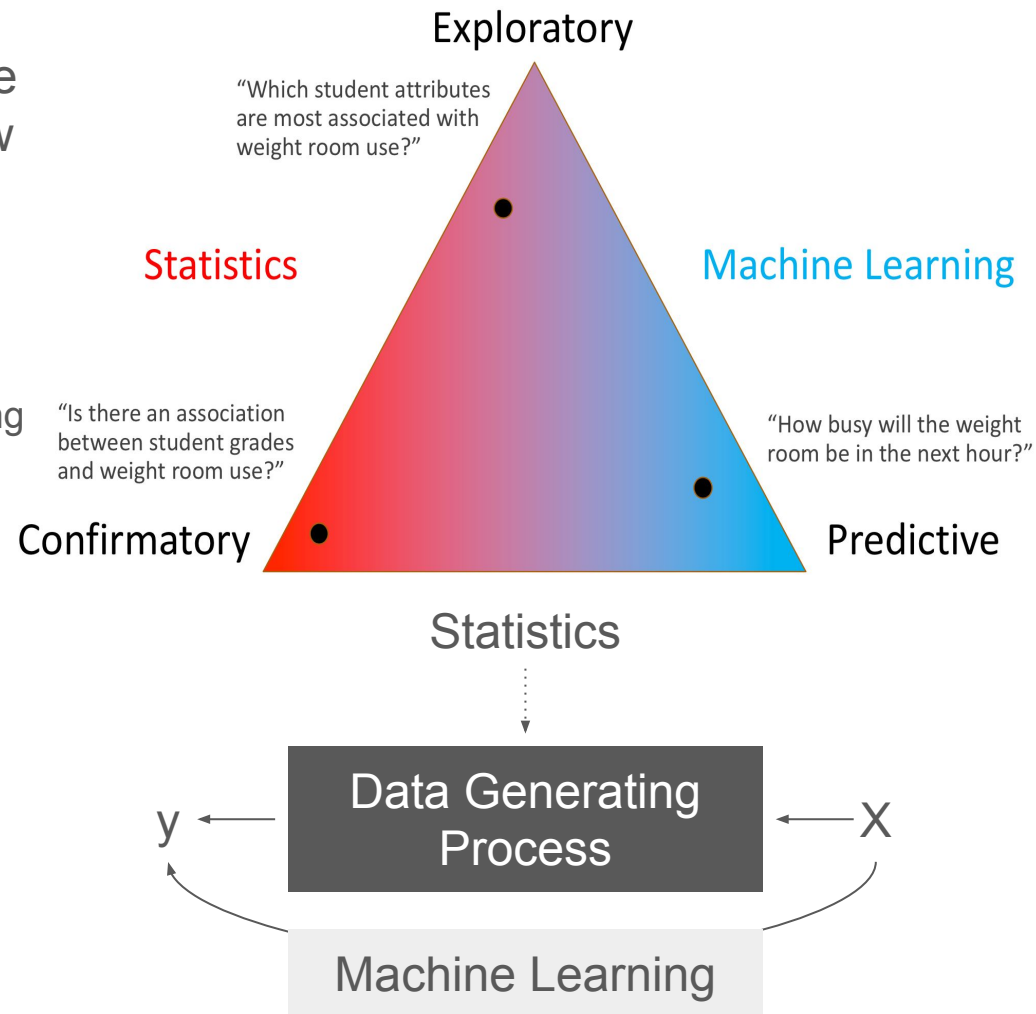
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- Alternative framing:
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- DS/ML Pitfalls (can be):
 - Less rigorous/principled
 - Prone to reinventing the wheel
- DS/ML Benefits (can be):
 - More flexible
 - Less prescriptive/intimidating



Why learn about Health Data Science
research?

Opportunity of Health Data Science

Benefits (and pitfalls!) of data science in general combined with:

- Huge amounts of health data

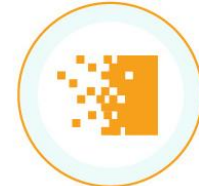
Growth in healthcare data

1 exabyte = 1 billion gigabytes

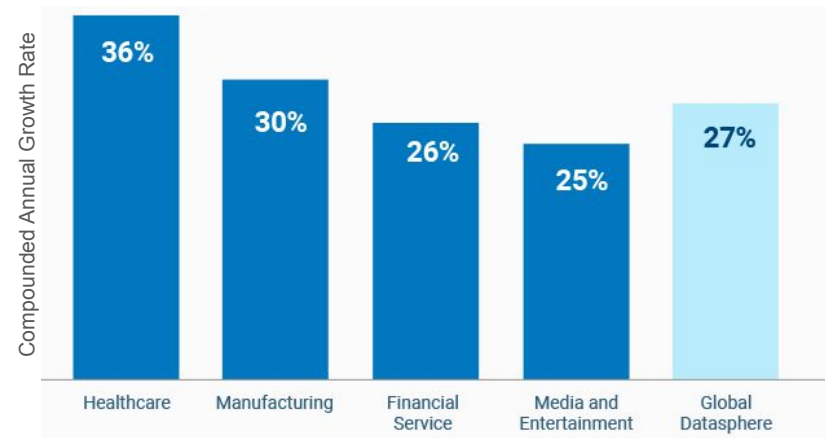
Source: Stanford Medicine 2017, IDC 2014



2013
153
EXABYTES



2020
2,314
EXABYTES



Opportunity of Health Data Science

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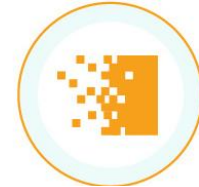
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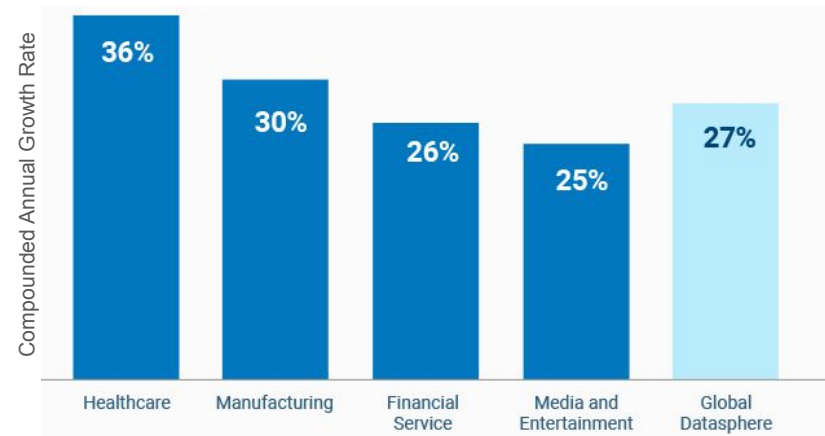
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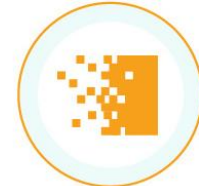
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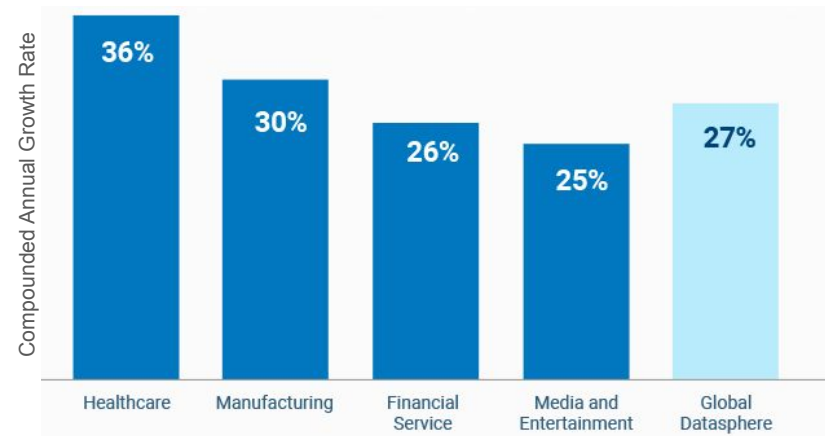
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Opportunity of Health Data Science

Benefits (and pitfalls!) of data science in general combined with:

- Huge amounts of health data
- Many **interesting** and **important problems**
- Many domain experts desperate for data-related help with these problems
- Relative few skilled data science practitioners

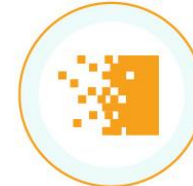
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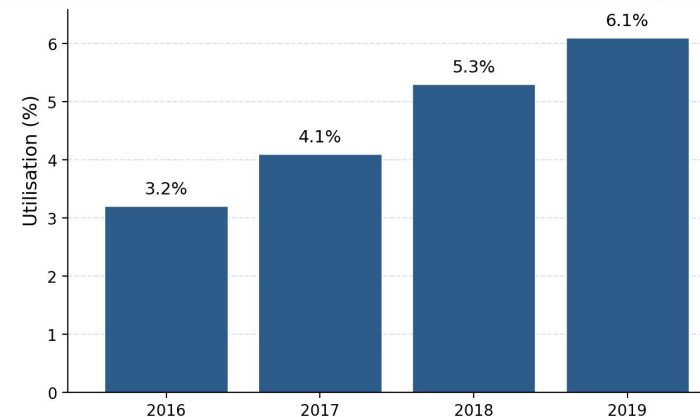
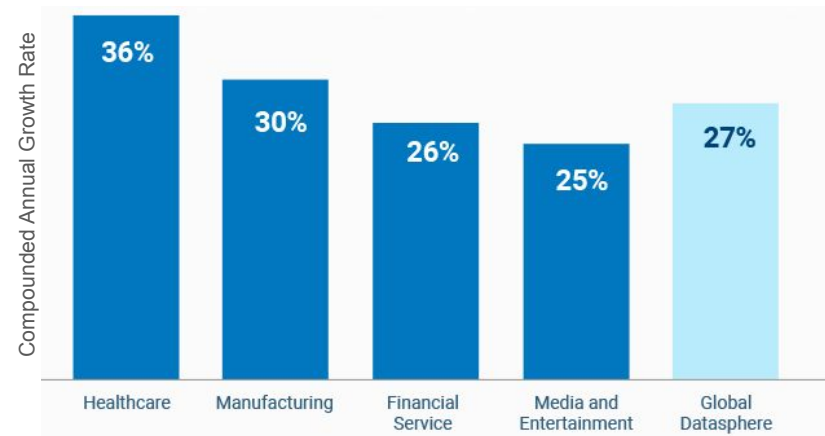
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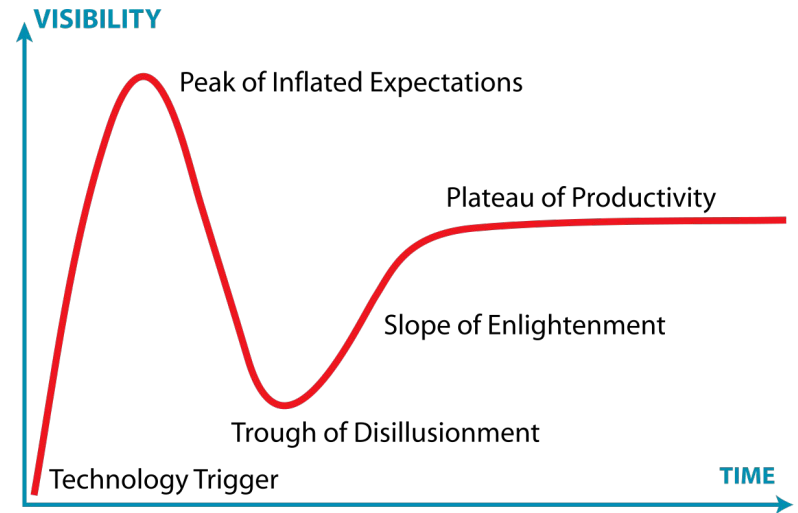
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Xuanwu Hospital: % of staff requesting EMR data for research (Li et al., JMIR Med Inform 2021)

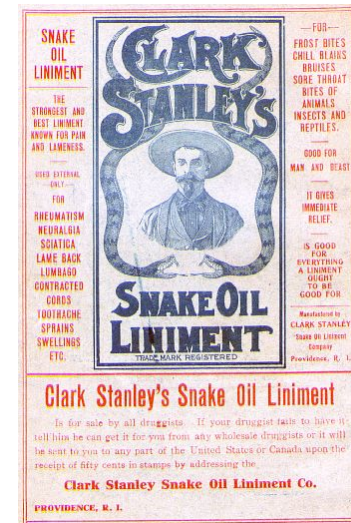
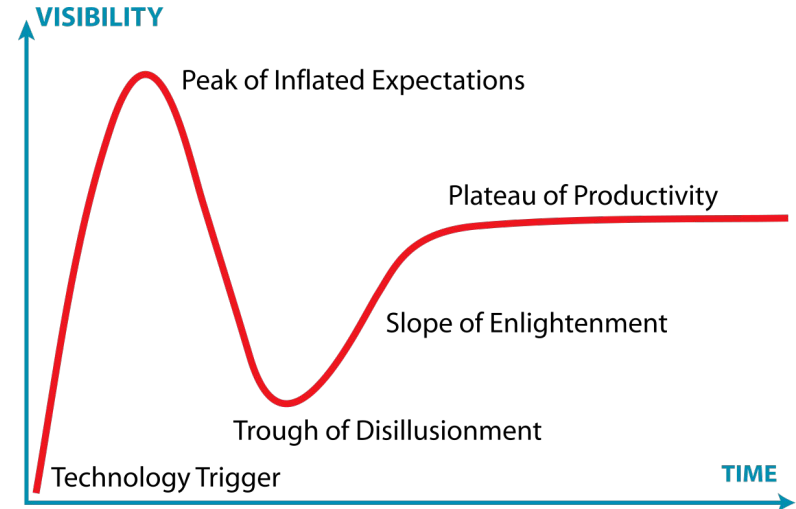
(Some) Challenges of Health Data Science

- Lots of hype



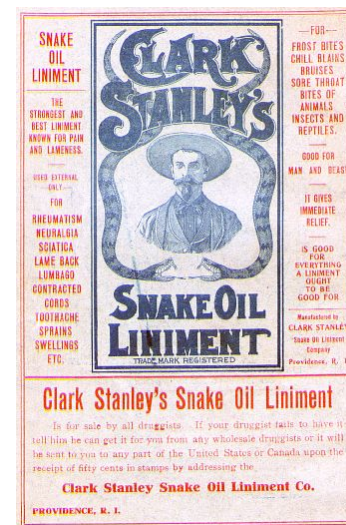
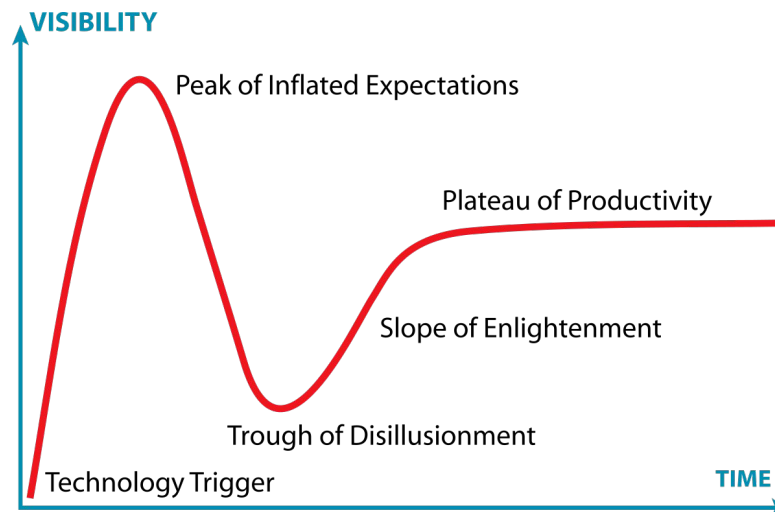
(Some) Challenges of Health Data Science

- Lots of hype
- Lots of grifters



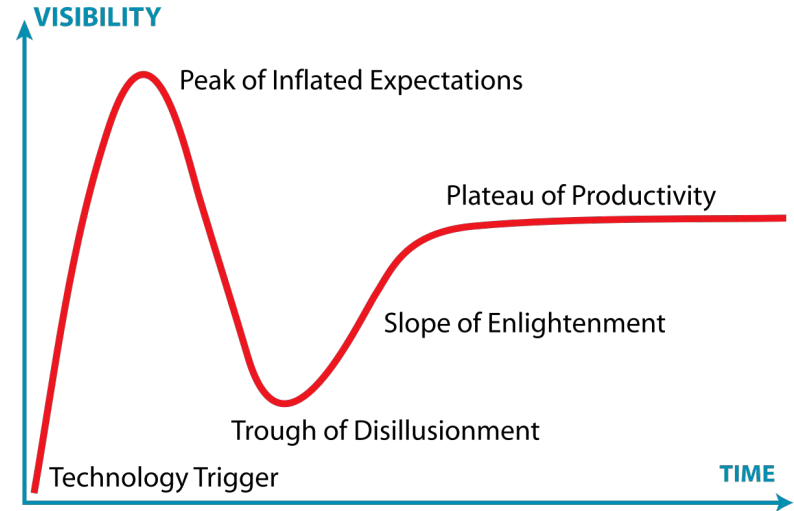
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- Lots of hype
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- Contextual/Metadata quality issues

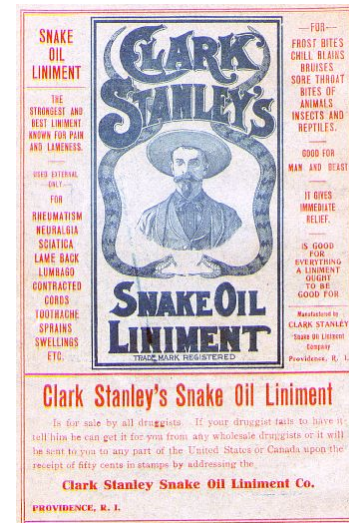


(Some) Challenges of Health Data Science

- Lots of hype
- Lots of grifters
- Data quality issues
- Contextual/Metadata quality issues
- Regulatory challenges
- Influence of US health system
- Ethical pitfalls
- Treatment to the mean
- Knowledge Translation and Operations: **Hard**

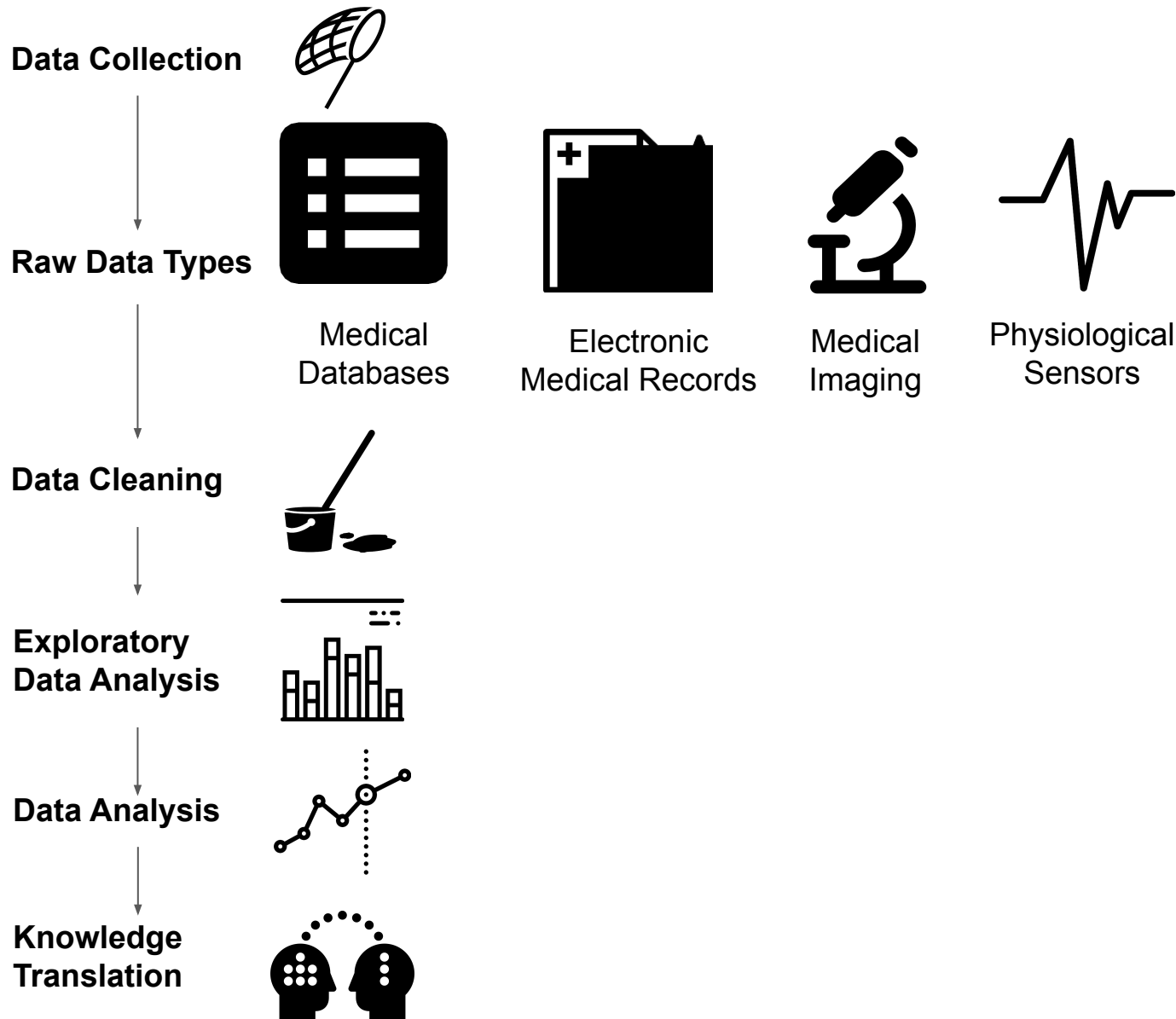


<https://www.r-bloggers.com/2019/08/new-course-learn-advanced-data-cleaning-in-r/>

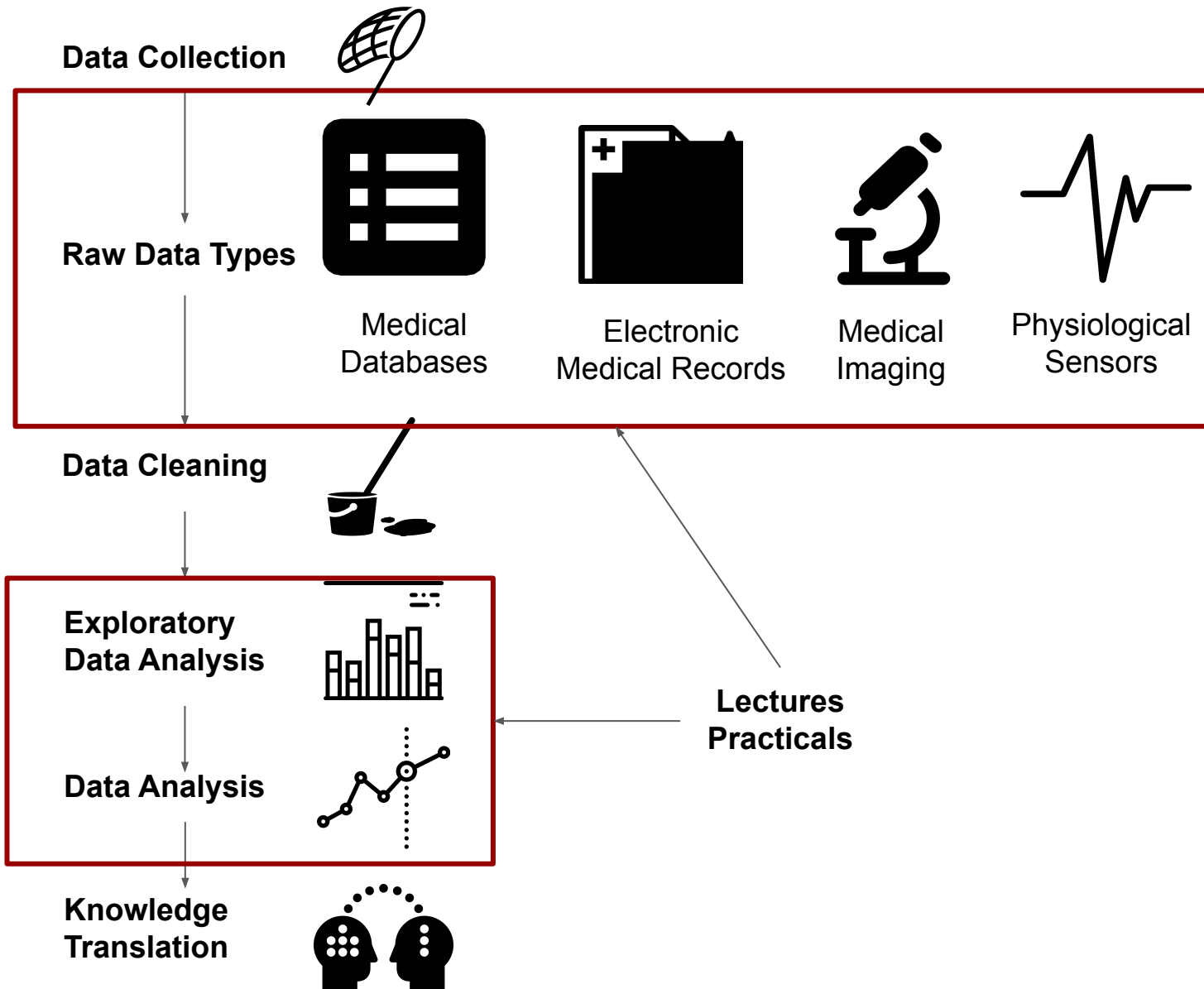


What parts of health data science will this course cover?

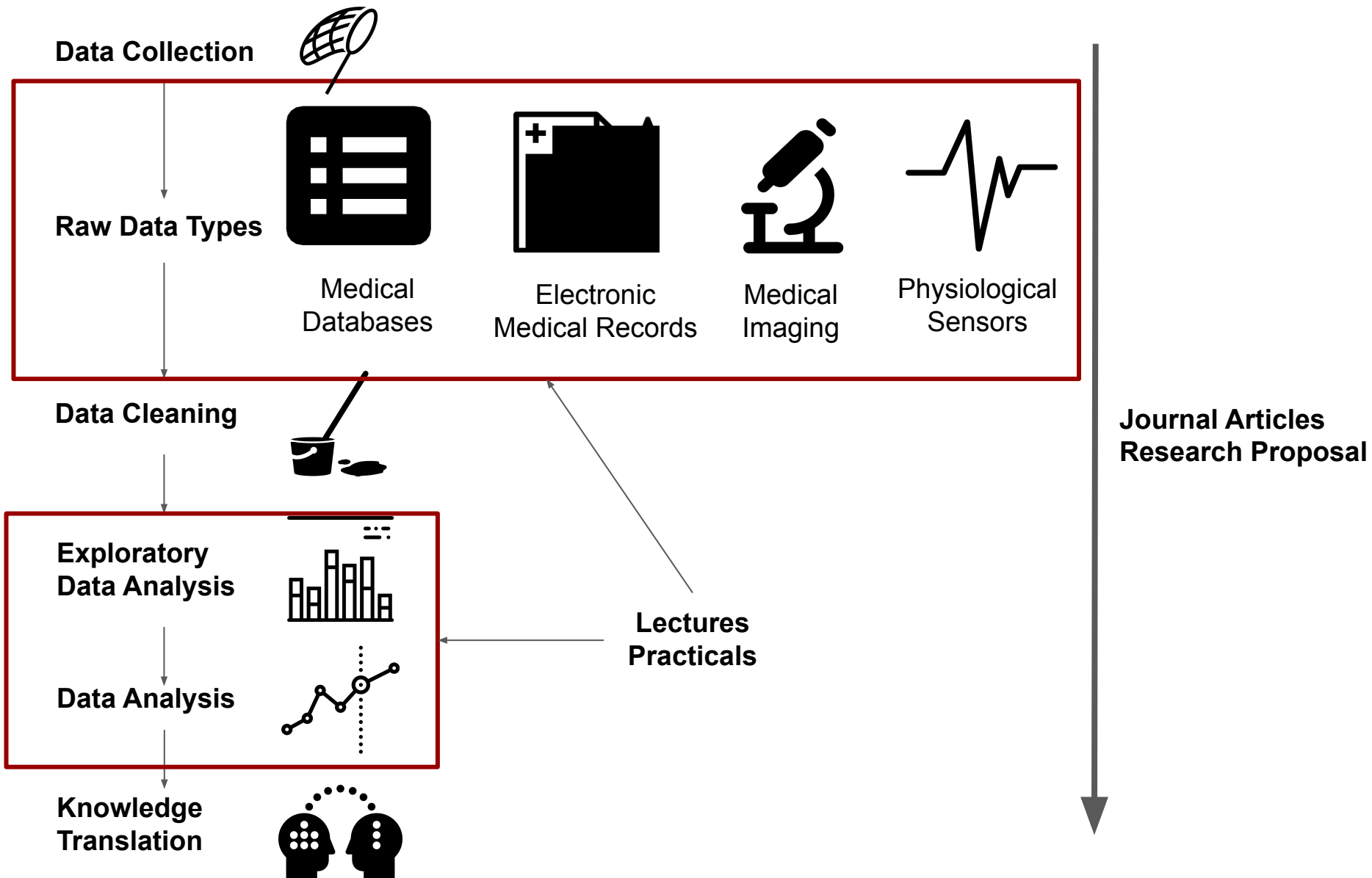
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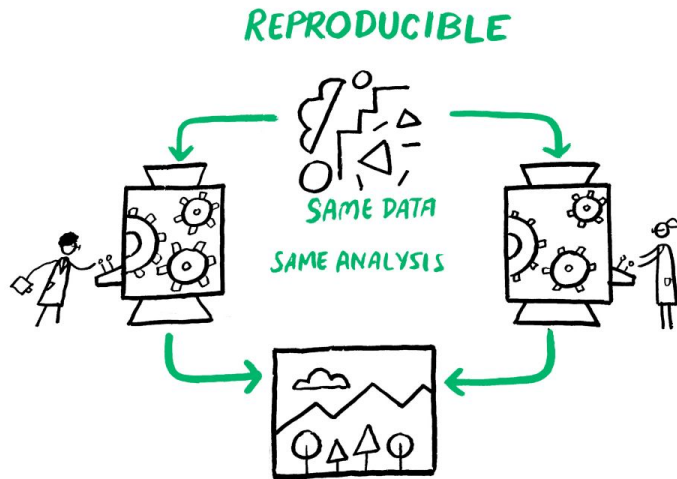
Let's take a 5 minute break!

Tools for Reproducible Health Data Science

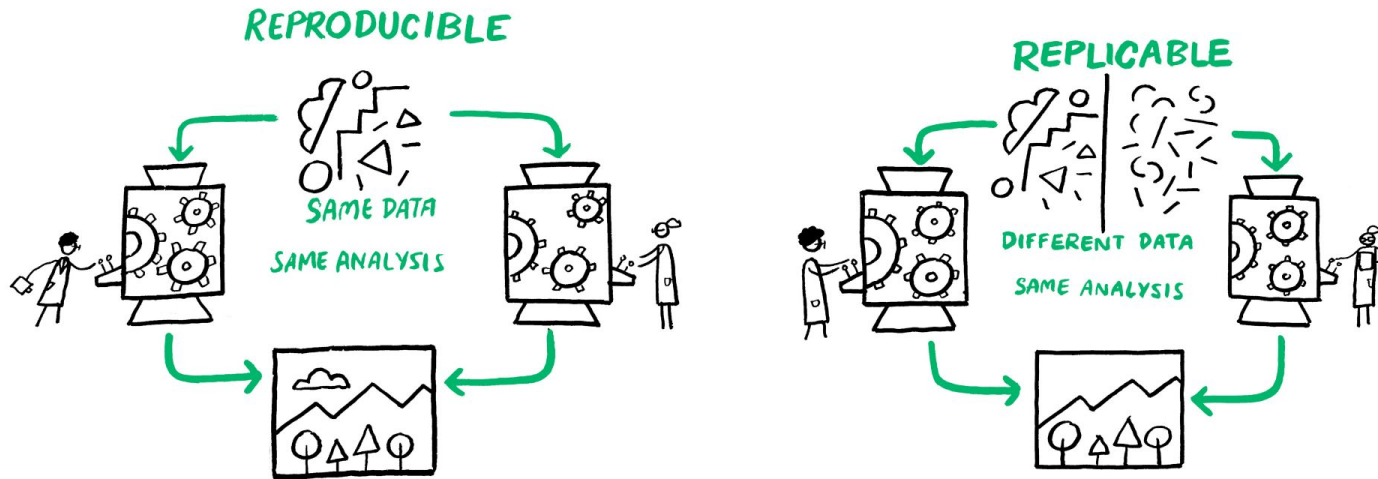
Rstudio, Rmarkdown, Git

Why do we care about reproducibility?

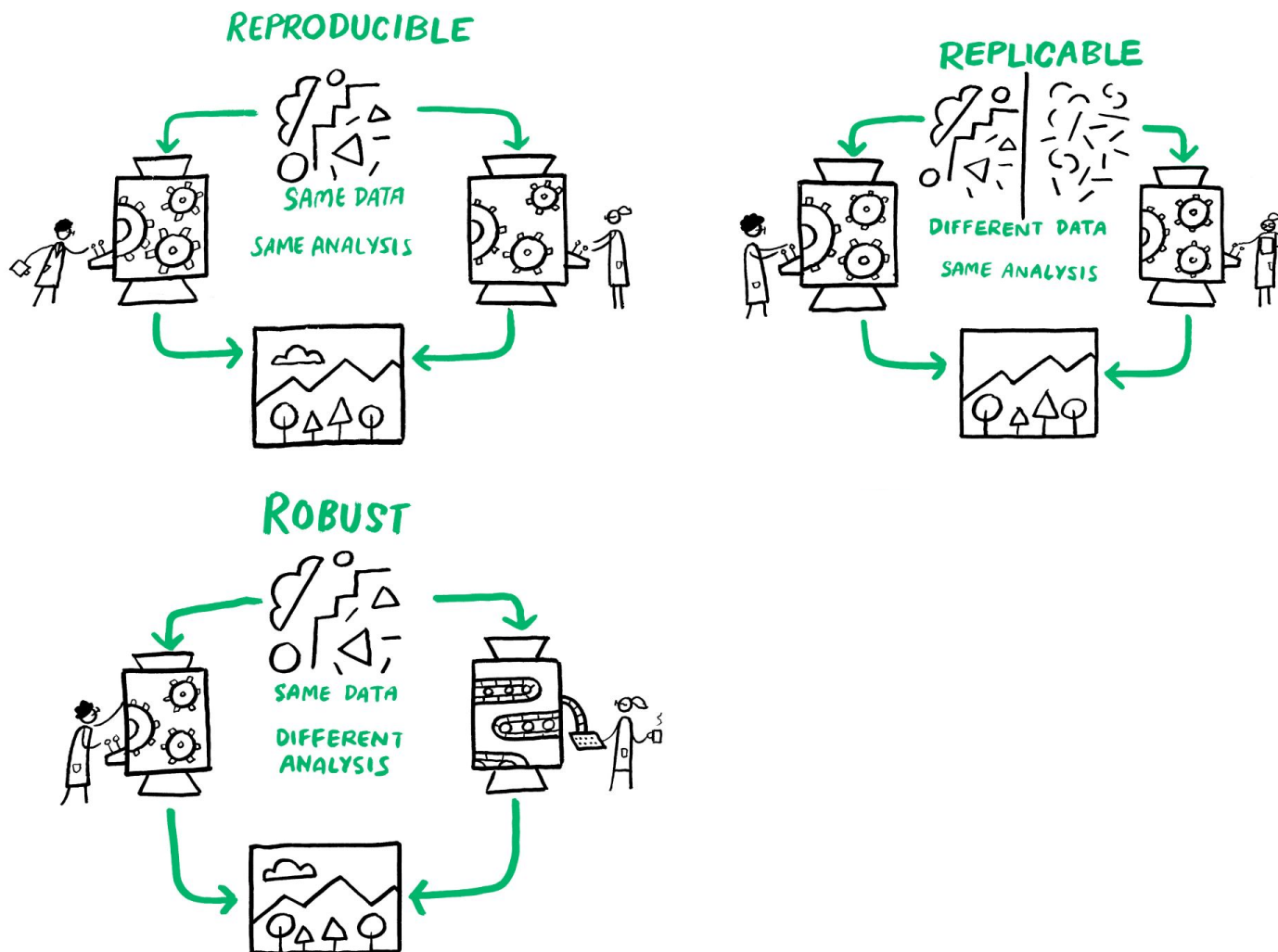
Reproducibility should be the bare minimum



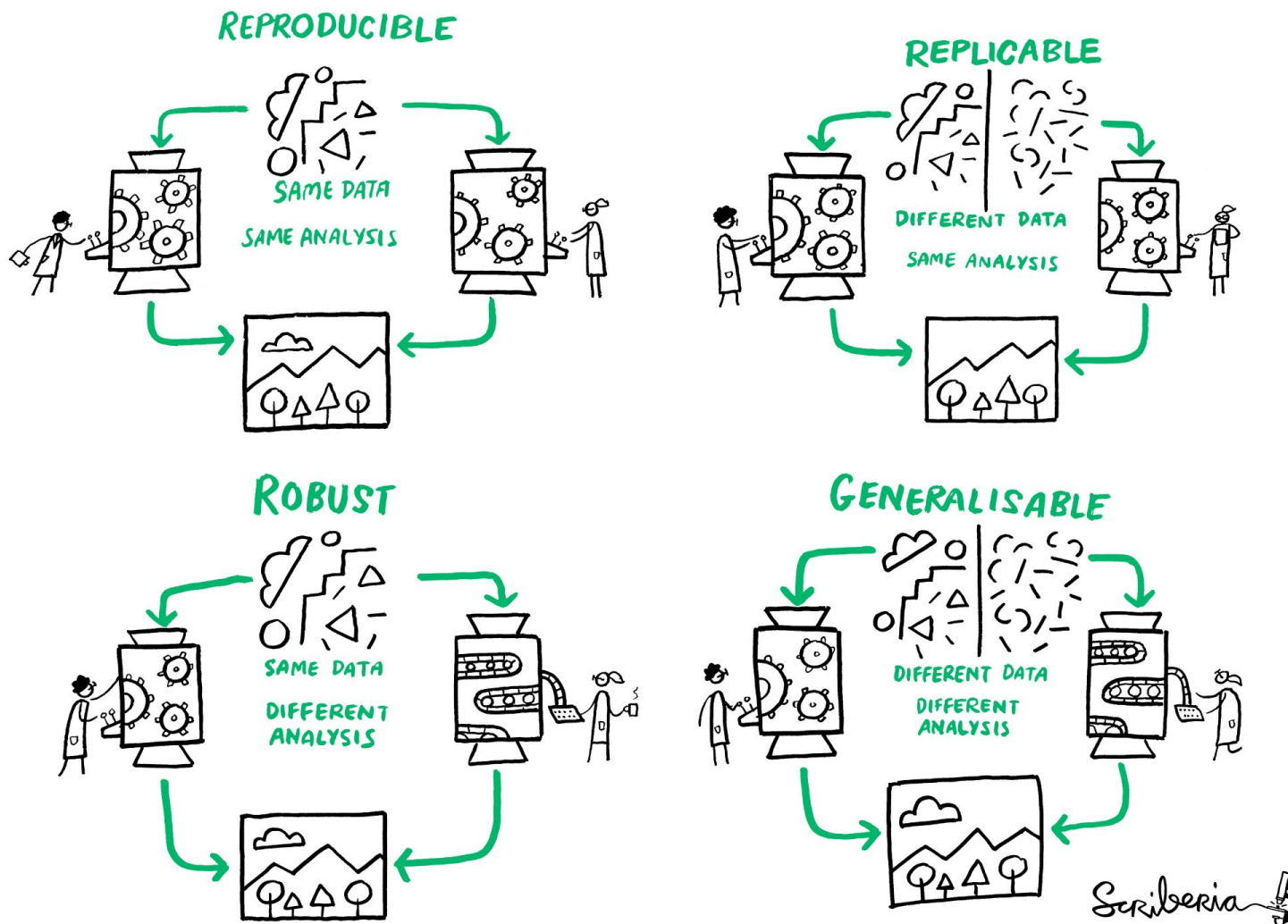
Reproducibility should be the bare minimum



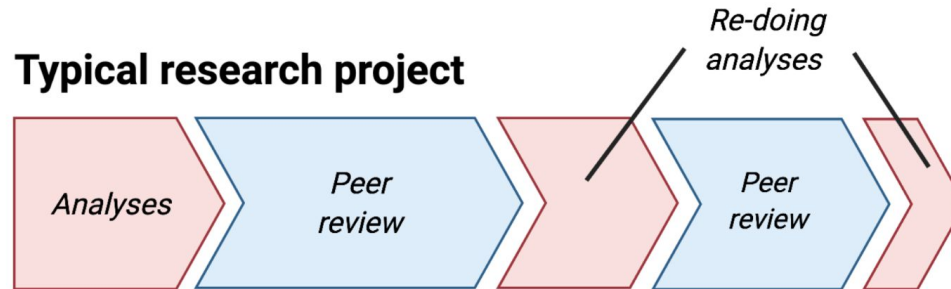
Reproducibility should be the bare minimum



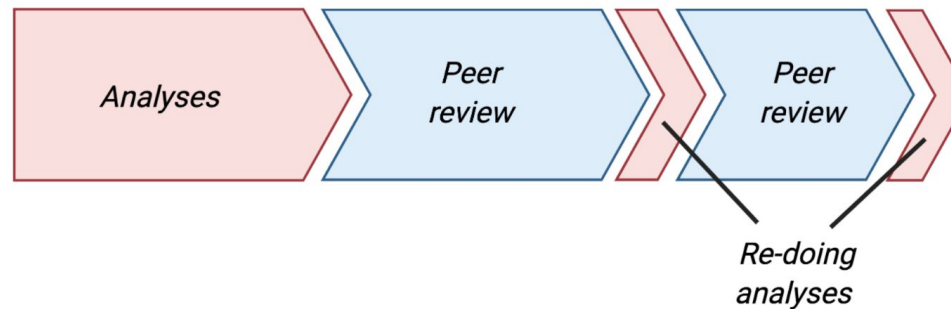
Reproducibility should be the bare minimum



Makes your own life easier



Research project using reproducible practices



@dsquintana

oliviergimenez.github.io/reproducible-science-workshop

What do we need to do to have reproducible research?

Reproducibility checklist

- Don't do anything by hand (even “one-off” tasks)

Reproducibility checklist

- Don't do anything by hand (even “one-off” tasks)
- Script every interaction with data:
 - Data collection
 - Moving data on your computer
 - Formatting datasets
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 - Exploratory data analysis
 - Main analyses
 - Report generation

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- Version control all data, code, and documentation
- Use a random seed
- Keep track of the exact version of every library/program you use
- Remove hard-coded/system specific paths

How do we actually do these things?

Choose a language that makes it easy to do most/all of your analysis

Choose a language that makes it easy to do most/all of your analysis



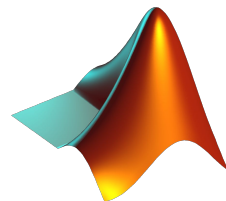
May 2026	May 2025	Change	Programming Language	Ratings	Change
1	1		 Python	19.98%	-5.37%
2	3	▲	 C	11.55%	+1.84%
3	4	▲	 Java	7.94%	-1.37%
4	2	▼	 C++	7.92%	-2.02%
5	5		 C#	5.41%	+1.19%
6	6		 JavaScript	3.08%	-0.60%
7	8	▲	 Visual Basic	2.90%	+0.28%
8	12	▲	 R	1.77%	+0.31%
9	10	▲	 SQL	1.57%	-0.33%
10	9	▼	 Delphi/Object Pascal	1.44%	-0.85%

<https://www.tiobe.com/tiobe-index/>

2026 consolidation toward R for statistical:

- MATLAB #20
- SAS #28
- Wolfram/Mathematica: #50-100
- S #102
- SPSS #112
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Use a data science focused IDE: Rstudio

The screenshot displays the RStudio interface with the following components:

- Source Editor:** Contains R code for loading the `nycflights13` dataset, creating a `daily` tibble, and plotting the number of flights per weekday.
- Environment:** Shows the `daily` object with 365 observations and 3 variables.
- Console:** Displays the execution of the code, showing the structure of the `daily` tibble and the first few rows of data.
- Plots:** A boxplot titled "Number of 2013 New York Flights Each Weekday" showing the distribution of flights for each day of the week.

R Code in Source Editor:

```
1 library(nycflights13) ## package containing flights dataset
2 library(lubridate)
3 library(dplyr)
4 library(ggplot2)
5
6 head(flights, n = 3)
7 daily <- flights %>%
8   mutate(date = make_date(year, month, day)) %>%
9   count(date) %>%
10  mutate(wday = wday(date, label = TRUE))
11 head(daily, n = 3)
12 ggplot(daily, aes(wday, n)) +
13   geom_boxplot(outlier.colour = "hotpink") +
14   labs(x = "Weekday", y = "Flights",
15        subtitle = "Number of 2013 New York Flights Each Weekday")
16
```

Console Output:

```
# A tibble: 3 x 19
  year month day dep_time sched_dep_time dep_delay arr_time sched_arr_time arr_delay carrier
<int> <int> <int> <int> <int> <dbl> <int> <int> <dbl> <chr>
1 2013 1 1 517 515 2 830 819 11 UA
2 2013 1 1 533 529 4 850 830 20 UA
3 2013 1 1 542 540 2 923 850 33 AA
# ... with 9 more variables: flight <int>, tailnum <chr>, origin <chr>, dest <chr>, air_time <dbl>,
# distance <dbl>, hour <dbl>, minute <dbl>, time_hour <dtm>
> daily <- flights %>%
+   mutate(date = make_date(year, month, day)) %>%
+   count(date) %>%
+   mutate(wday = wday(date, label = TRUE))
> head(daily, n = 3)
# A tibble: 3 x 3
  date n wday
<date> <int> <ord>
1 2013-01-01 842 Tue
2 2013-01-02 943 Wed
3 2013-01-03 914 Thu
> ggplot(daily, aes(wday, n)) +
+   geom_boxplot(outlier.colour = "hotpink") +
+   labs(x = "Weekday", y = "Flights",
+        subtitle = "Number of 2013 New York Flights Each Weekday")
>
```

Boxplot Data Summary:

Weekday	Min	Q1	Median	Q3	Max	Outliers
Sun	880	900	905	915	930	720, 730, 780, 810, 840, 850, 860, 870, 880, 890, 900, 910, 920, 930, 940, 950, 960, 970, 980, 990, 1000
Mon	910	920	930	940	950	900, 910, 920, 930, 940, 950, 960, 970, 980, 990, 1000
Tue	900	910	920	930	940	880, 890, 900, 910, 920, 930, 940, 950, 960, 970, 980, 990, 1000
Wed	910	920	930	940	950	900, 910, 920, 930, 940, 950, 960, 970, 980, 990, 1000
Thu	910	920	930	940	950	900, 910, 920, 930, 940, 950, 960, 970, 980, 990, 1000
Fri	910	920	930	940	950	900, 910, 920, 930, 940, 950, 960, 970, 980, 990, 1000
Sat	680	690	700	710	720	730, 740, 750, 760, 770, 780, 790, 800, 810, 820, 830, 840, 850, 860, 870, 880, 890, 900, 910, 920, 930, 940, 950, 960, 970, 980, 990, 1000

.Rproj file
here()

set.seed()
sessionInfo()

Use notebooks to document analyses: Rmarkdown/Quarto

~\Documents\rmarkdown - gh-pages - RStudio

9-notebook.Rmd

```
1 ---
2 title: "Viridis Notebook"
3 output: html_notebook
4 ---
5
6 ```{r include = FALSE}
7 library(viridis)
8 ```
9
10 The code below demonstrates two color palettes in the
11 [viridis](https://github.com/sjmgarnier/viridis) package. Each
12 plot displays a contour map of the Maunga Whau volcano in
13 Auckland, New Zealand.
14
15 ## Viridis colors
16 ```{r}
17 image(volcano, col = viridis(200))
18 ```
```

3:22 Viridis Notebook R Markdown

Environment History Build Git

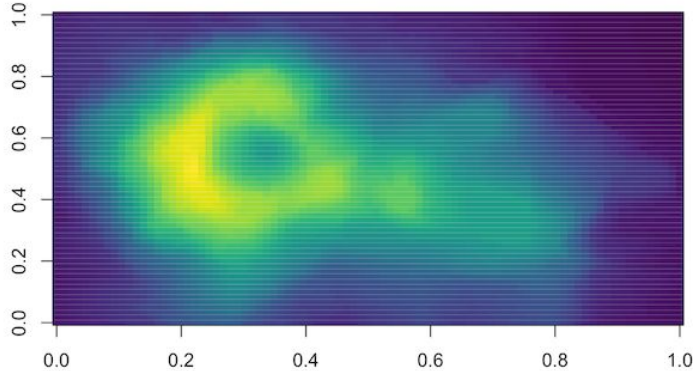
Files Plots Packages Help Viewer

Viridis Notebook

The code below demonstrates two color palettes in the [viridis](https://github.com/sjmgarnier/viridis) package. Each plot displays a contour map of the Maunga Whau volcano in Auckland, New Zealand.

Viridis colors

```
image(volcano, col = viridis(200))
```



Magma colors

Use notebooks to document analyses: Rmarkdown/Quarto

settings). Therefore, from this time onward, case counts are likely underestimated and the sequenced virus diversity is not necessarily representative of the virus circulating in the overall population.

BC AB SK MB ON QC NS NB NL

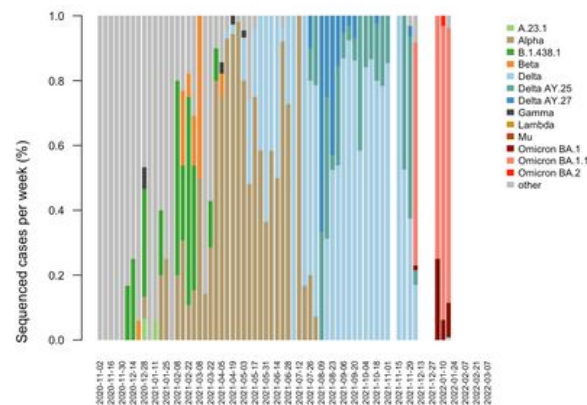
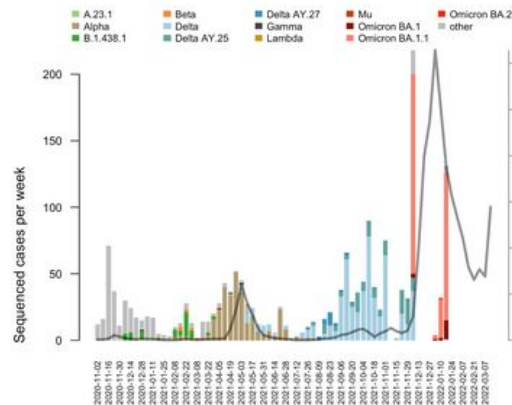
Nova Scotia

Additional up-to-date COVID data for this province can be found here:

<https://experience.arcgis.com/experience/204d6ed723244dfbb763ca3f913c5cad>

Hide

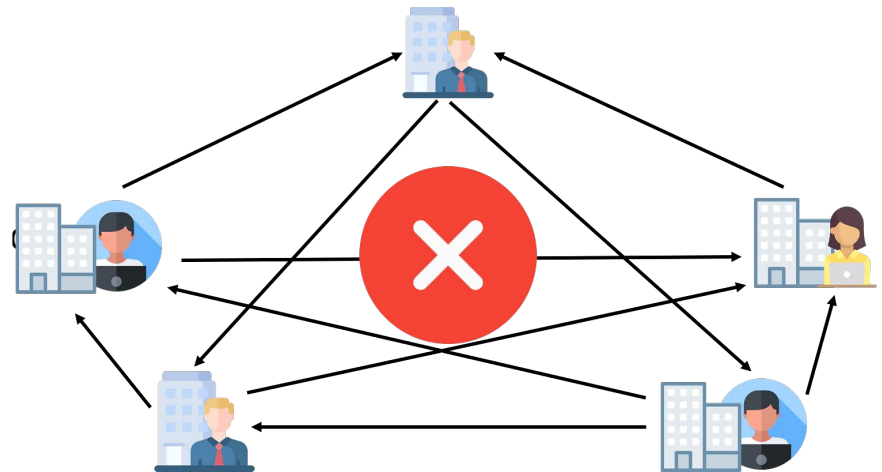
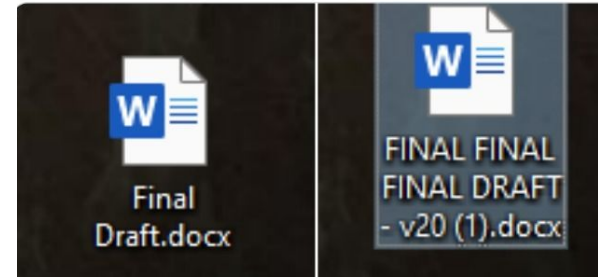
```
plot.variants(region='Nova Scotia')
plot.variants(region='Nova Scotia', scaled=T)
```



<https://covarr-net.github.io/duotang/duotang.html#>

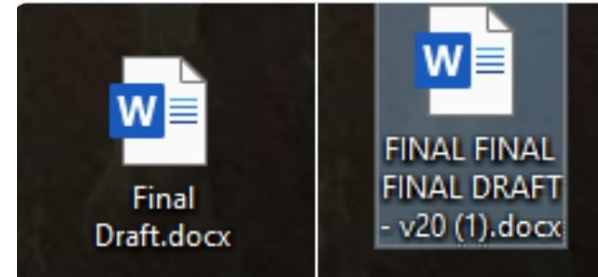
Use standard version control systems

- Ever had a nightmare of versioning even when just you?
- Add more people and the chaos grows exponentially!



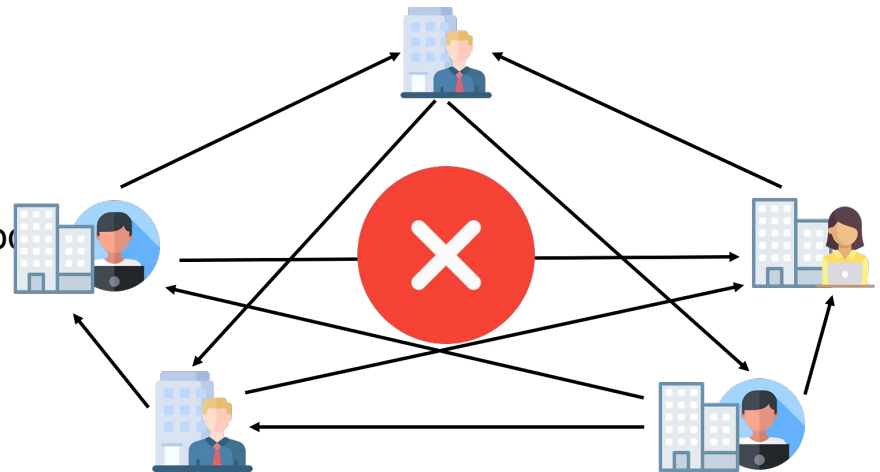
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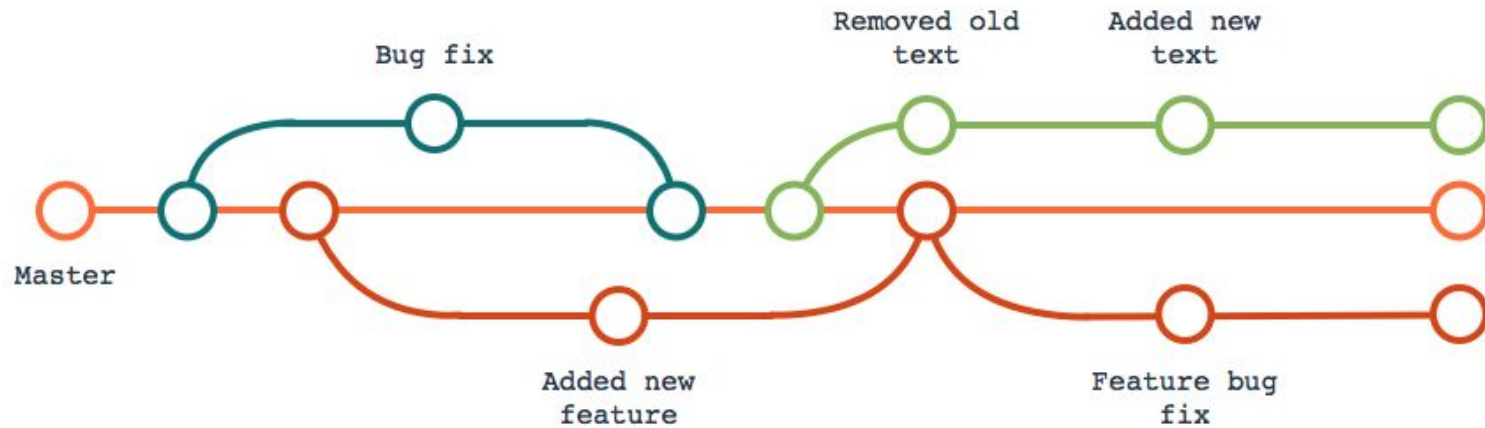


Version control let's you:

- Revert mistakes
- Acts as a comprehensive backup
- Let's you maintain multiple versions of your analysis
- Let's you compare different versions of your code
- Track down the who/what broke the analysis
- Work out why you did something in the past
- Build on someone else's work
- Share your own work
- Experiment without risk

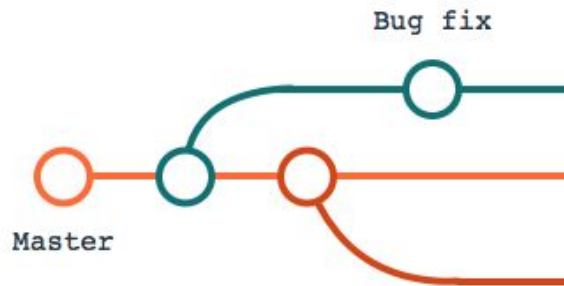


Git Version Control

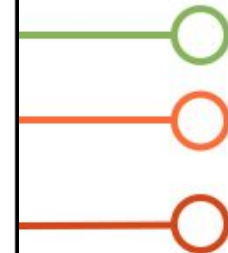


- Most popular
- Decentralised
- Designed for
- GitLab/GitHub Services

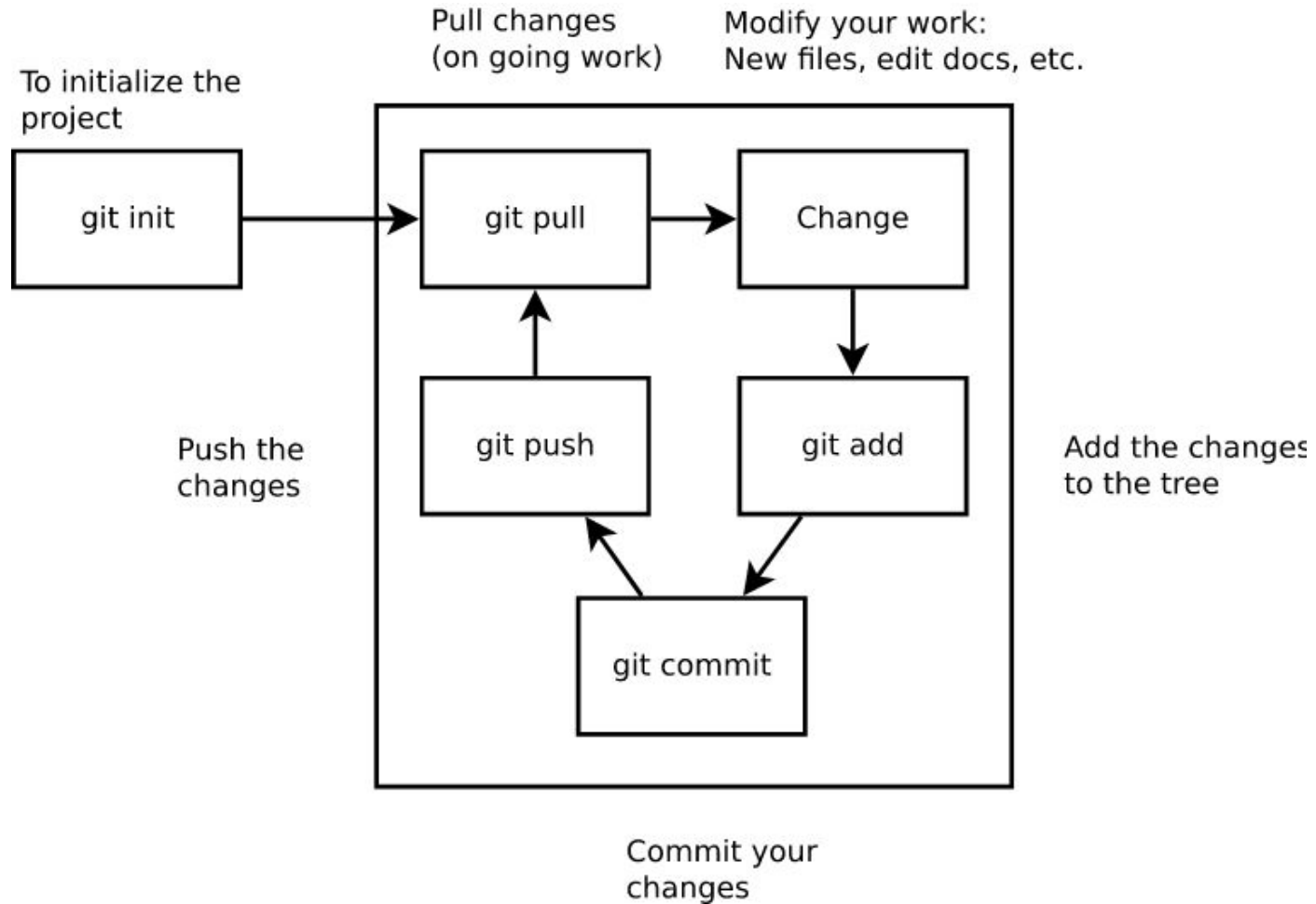
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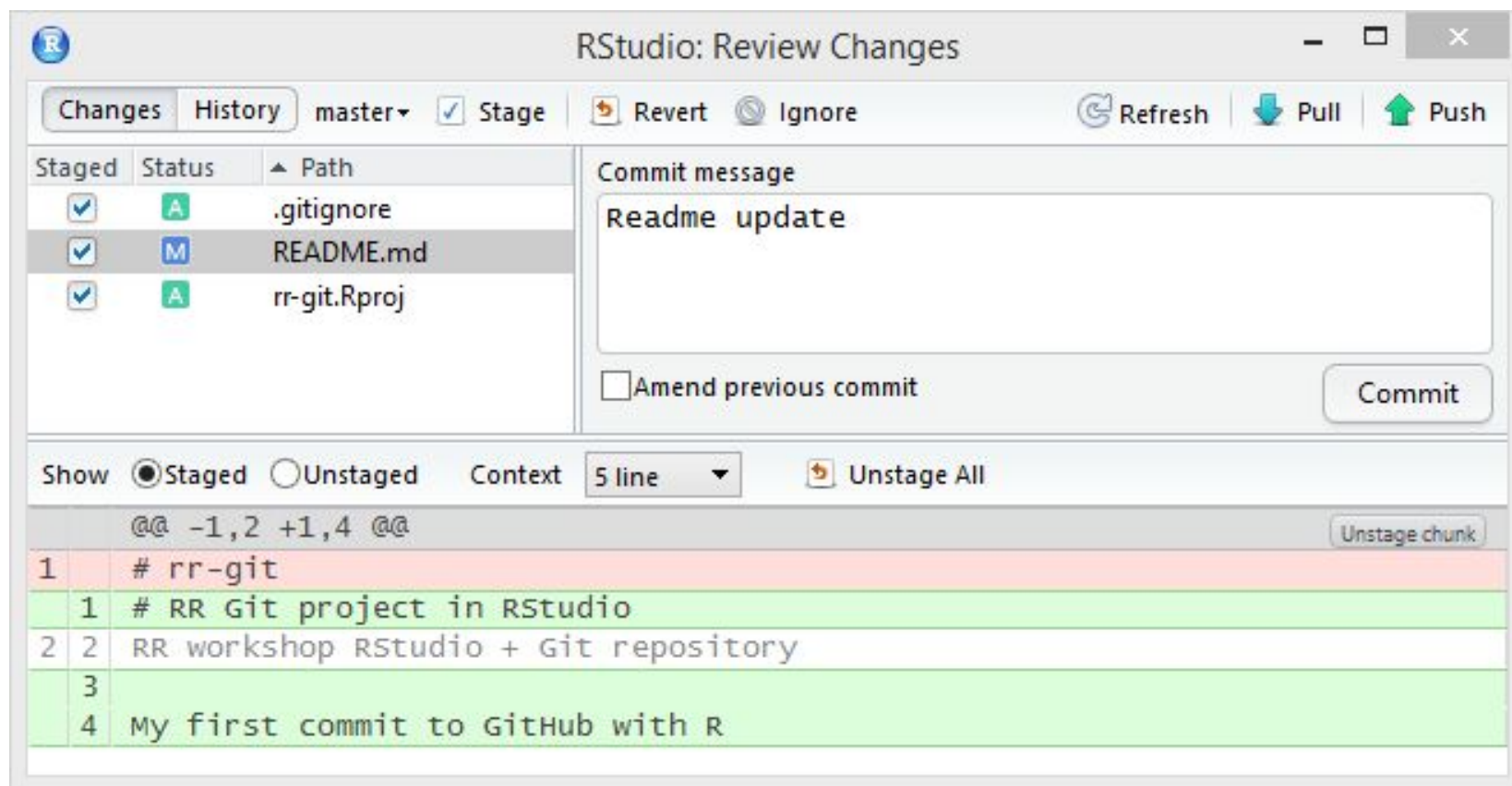
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Git Workflow

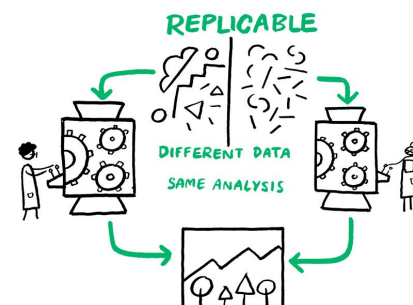
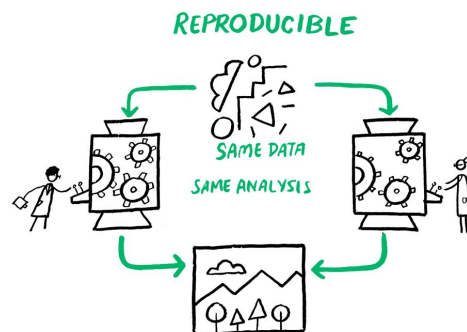


Git is integrated into Rstudio!

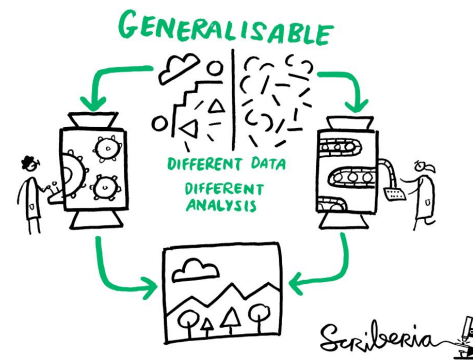
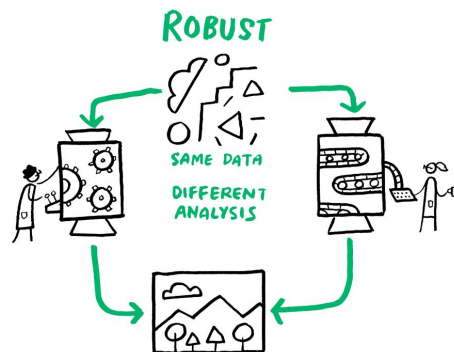


Combine Git+Rmd Notebooks for Reproducibility

1. Add analysis to notebook
2. Add changes to git
3. Find out you made a mistake
4. Revert changes



1. Share notebook with collaborator
2. They make changes
3. You make changes
4. Merge changes into single analysis



Scriberia

Summary

- Overview of course: Database/EMR/Imaging/Signal
- Main assessments: practicals, journal article presentations, research proposal
- Data science is statistics with an EDA/Inductive/Data-focused Spin
- Health Data Science is a massive and growing area with lots of opportunity and challenges
- R is a powerful and useful tool for health data science
- Reproducibility is vital to good ~~health-data~~ science
- Rstudio, Rmarkdown notebooks and Git based version control facilitate that reproducibility

Friday's Practical

- Will go over the practical use of R, Rstudio, Rmd Notebooks, Git
- Try and install rstudio, git, and rmarkdown beforehand.
- 1st practical will not contribute to your course grade

Wednesday's Journal Articles

A Beginner's Guide to Conducting Reproducible Research

Jesse M. Alston, Jessica A. Rick First published: 15 January 2021 <https://doi.org/10.1002/bes2.1801>

Challenges to the Reproducibility of Machine Learning Models in Health Care

Andrew L. Beam, Arjun K. Manrai, Marzyeh Ghassemi <https://doi.org/10.1001/jama.2019.20866>

JAMA • 28 January 2020 • Volume 323, Number 4

Reproducibility in machine learning for health research: Still a ways to go

[Matthew B. A. McDermott](#) [Shirly Wang](#) [Nikki Marinsek](#) [Rajesh Ranganath](#) [Luca Foschini](#) [Marzyeh Ghassemi](#)

Science Translational Medicine • 24 Mar 2021 • Vol 13, Issue 586 • [DOI: 10.1126/scitranslmed.abb1655](https://doi.org/10.1126/scitranslmed.abb1655)